Supplementary Information

DeSurv identifies a prognostic tumor–stroma architecture in pancreatic cancer through survival-supervised matrix factorization

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Contents

List of Figures **2**

List of Tables **3**

I Supplementary Methods **4**

1 Model details **4**

2 Optimization Algorithm **5**

2.1 Block coordinate descent scheme 5

2.2 Update for H 5

2.3 Update for W 5

2.4 Update for β 8

2.5 Normalization of W 9

3 Supervision on W versus H **9**

4 Cross-validation procedure **10**

5 Bayesian Optimization **11**

5.1 Consensus initialization for the final model 13

6 PDAC Datasets **13**

7 Survival Analysis **14**

7.1 Independence of discovery and validation 15

7.2 Cohort handling in the training objective 15

8 Gene Program Correlation Analysis **15**

9 Reconstruction and Survival Contribution per Factor **16**

10 Factor Correspondence Analysis **17**

11 Analysis scripts and cached objects **17**

II Supplementary Results 18

12 Cross-method reconstructability and reconstruction fidelity **18**

13 The D1 axis is not specific to DeSurv **18**

14 Robustness to tuned supervised comparators and compartment attribution **18**

15 Proper-scoring sensitivity of the external validation **19**

16 Proportional-hazards diagnostic **20**

17 Between-cohort heterogeneity and C-index confidence intervals **20**

17.1 Leave-one-dataset-out sensitivity of the pooled hazard ratio 21

18 Variance partition of the D1 score **21**

19 Adjustment for clinical covariates and tumor purity **21**

20 Empirical optimization stability across random initializations **22**

21 Stability of the recovered gene programs **22**

22 Simulation design and data generation **23**

23 Simulation results under null and mixed scenarios **24**

24 Cross-validated C-index across factorization rank **25**

25 Standard NMF rank selection heuristics **26**

26 Factor structure of the rank-optimized unsupervised control ($k = 7$) **26**

26.1 Reference-program coverage, edges and multiplicity 26

26.2 Stromal resolution: which reference programs are recovered from bulk 27

26.3 Matched-budget unsupervised control at the same rank 27

27 Supplementary Data: factor-specific gene lists **28**

28 Translational transportability into independent treated PDAC cohorts **29**

28.1 Prognostic transportability of D1 in treated metastatic disease 29

28.2 Treatment-associated change in D2 29

28.3 Caveats 29

SI References **30**

List of Figures

S1 Bayesian-optimization tuning surface for DeSurv in the PDAC training cohort (TCGA + CPTAC). Gaussian-process–predicted mean cross-validated C-index over factorization rank (k) and supervision strength (α; $\alpha = 0$ corresponds to the unsupervised limit). The response surface shows a broad region of high cross-validated performance at intermediate supervision and moderate rank, a second high region at $k \geq 9$ and $\alpha \approx 0.2$–0.3, and discrimination degrading toward the unsupervised ($\alpha = 0$) boundary. The surface is a two-dimensional slice of the fitted five-parameter Gaussian process through the Bayesian-optimization-selected values of the top-gene count, ridge penalty and elastic-net mixing parameter (n_{top}, λ, ξ), not a marginal or profile surface over those hyperparameters. 32

S2 Correspondence of supervised scores and the first unsupervised principal component with D1 in the training cohort. (a) Absolute correlation (across training-cohort patients) of each method's leading direction with the DeSurv D1 score, in the training cohort: supervised principal components, Cox partial least squares, and sparse Cox are each strongly correlated with D1 (green), whereas the leading unsupervised principal component is only weakly correlated (grey). (b) Sample-level agreement with D1 for Cox-PLS (supervised, green) versus the first unsupervised principal component (grey). 33

S3 DeSurv performance under null and mixed simulation scenarios. (a,b) Null scenario ($\beta = 0$): both methods yield C-index ≈ 0.5 (a) and comparable rank distributions (b), so both methods show no out-of-sample prognostic discrimination under the null; precision is omitted as no true survival genes exist. (c–f) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (c) C-index is modestly higher for DeSurv. (d) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (e) The matched factor receives a larger fitted Cox coefficient $|\beta|$ under DeSurv. (f) Selected-rank distributions. DeSurv (blue); NMF (red, $\alpha = 0$); both Bayesian-optimized. 34

S4 Training (CV) and external validation C-index as a function of factorization rank k for DeSurv (blue) and its unsupervised limit (red, $\alpha = 0$). Shaded ribbons show ± 1 SE. For each k, the best α was selected by maximizing the mean CV C-index. Validation C-index is a single Harrell concordance computed on the de-duplicated external patients of the four validation cohorts pooled together, so concordance pairs cross cohorts, with the ribbon giving plus or minus one standard error from that single concordance estimate. This descriptive quantity differs from the primary combined external C-index reported in the main text, which is an event-weighted mean of four cohort-specific C-indices, and the two are not directly comparable. (a) Results using $n_{\text{top}} = 270$ top genes per factor ($\lambda = 0.349$, $\xi = 0.056$). (b) Results retaining all genes ($n_{\text{top}} = \text{ALL}$; $\lambda = 0.349$, $\xi = 0.056$). 35

S5 Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (a) Reconstruction residuals as a function of factorization rank k. (b) Cophenetic correlation coefficient as a function of k. (c) Mean silhouette width of the basis, coefficient and consensus matrices as a function of k. These criteria yield inconsistent guidance, illustrating the ambiguity of rank selection under unsupervised criteria; the main text selects rank on cross-validated prognostic performance. 36

S6 Gene-program correspondence for the rank-optimized unsupervised control ($k = 7$, BO-selected rank at $\alpha = 0$), computed with the same Spearman-correlation analysis and reference programs as main-text Fig. 2a,b (top 50 genes per factor; reference programs with $ho > 0.2$ for at least one factor are shown). Compared with the matched-rank $k = 3$ solution, the $k = 7$ solution shows broader correspondence to established PDAC programs, with some programs represented by more than one factor. Supplementary Fig. S7 quantifies the accompanying changes in coverage and multiplicity. 37

S7 Reference-program coverage, edges and multiplicity by factorization rank k, for the unsupervised ($\alpha = 0$) and supervised (best α by cross-validated C-index) curves, computed from the same factor-to-reference-program correspondence analysis as Fig. S6 and main-text Fig. 2a,b (top 50 genes per factor, $ho > 0.2$ threshold, same reference-program universe). Rank-wise points are single fits at fixed λ, ξ and n_{top}, not the 100-restart consensus fits used elsewhere; markers denote the three final-model anchors (the consensus fits for DeSurv $k = 3$ and the rank-optimized unsupervised control $k = 7$, and the best-of-30-run `NMF::nmf` fit at matched rank $k = 3$). 38

List of Tables

S1 Where survival supervision enters, and what a new sample requires, for DeSurv and five comparators. Cells for DeSurv, standard NMF, supervised principal components and penalized Cox regression are taken from the Methods of this manuscript; CoxNMF cells from its published objective function and SurvNMF cells from its full text; both works are cited in the text of this section. "Not described in the source" means the cited work does not state the behavior; no other cell is inferred. The table records architecture and does not rank the methods. Note: CoxNMF does not describe a procedure for scoring new samples. The implication that sample-side loadings would need to be obtained for new samples follows from its published objective and is an inference, not a reported implementation. 39

S2 Cross-validation of the final DeSurv hyperparameter configuration. The overall-best $k = 7$ configuration, the best $k = 3$ configuration identified during Bayesian optimization, and the final fitted $k = 3$ configuration were re-evaluated using the same DeSurv software build and the Bayesian-optimization cross-validation protocol: five folds constructed within the training data and stratified jointly by event status and cohort (seed 123), 30 random initializations per configuration and fold, and the same preprocessing, stopping tolerance, iteration cap and n_{top}-based scoring procedure. Values are mean Harrell C-index across folds with fold-level standard error. The cached Bayesian-optimization history was generated with an earlier DeSurv installation whose package version was not recorded; the best $k = 3$ configuration has C-index 0.647 under the current build versus 0.646 in the historical optimization object. The same-build values shown here are therefore used for direct numerical comparison. The final configuration lies 0.002 below the recomputed one-standard-error threshold of 0.644 (0.655 minus its fold-level standard error). 40

S3 Summary of PDAC cohorts used in this study. N denotes the number of samples after all quality-control filters (histology, survival validity, dataset-specific exclusions). Patient-level metadata: OS = overall survival (time and event indicator); PurIST = tumor subtype classification; DeCAF = CAF subtype classification. Expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study. Events denotes the number of overall-survival events (deaths) with the corresponding event rate; outcome frequency was lower and more homogeneous in the training cohorts (50–52%) than across the validation cohorts (60–90%, pooled 67%), a case-mix difference relevant to interpreting external-validation performance. 41

S4 The 45-program reference universe used for every factor-to-program correspondence analysis (main-text Fig. 2a,b; Supplementary Figs. S6 and S7), generated from the same renamed and de-duplicated object the analyses read. Displayed alias is the row label used in the figures; Genes is the size of the published list; In analysis set is the number of those genes among the 1,970 analyzed genes, over which the Spearman correlation is computed. The SCISSORS restCAF and proCAF rows are the SCISSORS iCAF and myCAF signatures under DeCAF nomenclature. The two Puleo basal-like rows are different programs: the whole-tumor pure basal-like component and the tumor-compartment basal-like signature. 42

S5 Index of analysis scripts. Scripts are numbered as in the code repository; helper functions shared by several scripts (de-duplication of PACA-AU patients, validation scoring, elbow selection) are called from the scripts listed. Section numbers refer to this document. 43

S6 Cross-method projection R^2: how reconstructable each factor's gene-loading vector is from the other method's full basis (OLS R^2). DeSurv retains the bulk of the unsupervised structure (NMF N1 and N3 carry over at $R^2 \geq 0.88$) and reorganizes it: the survival-aligned D1 program is essentially unreconstructable from the NMF basis ($R^2 = 0.09$), while the exocrine NMF factor N2 is poorly reconstructed from the DeSurv basis ($R^2 = 0.38$). 44

S7 Whole-matrix reconstruction R^2 across the 1,970-gene analysis matrix for standard NMF versus DeSurv at $k = 3$. Two denominators are shown. The centered R^2 is the conventional definition, taken about the grand mean, and is the one to read as fraction of variation reconstructed. The uncentered R^2 divides by $\sum X^2$; because the analysis matrix holds within-sample ranks with grand mean $p/2$, the mean alone accounts for about 75% of that denominator, so uncentered values are high for any nonnegative low-rank fit and are reported for completeness. On the centered definition DeSurv gives up 6.0 percentage points of reconstruction relative to NMF (1.5 on the uncentered scale). 45

S8 Leave-one-dataset-out sensitivity of the pooled external-validation hazard ratio for the frozen DeSurv linear predictor. Each row refits the dataset-stratified Cox model on the de-duplicated patient-level table with the named validation dataset omitted; the first row is the full combined estimate reported in main-text Table 1. Hazard ratios are per pooled standard deviation of the linear predictor, recomputed within each subset, the same convention as the headline pooled estimate and not the within-cohort convention of Table S11. Puleo contributes 288 of the 570 unique patients, which is why this sensitivity analysis is reported. 46

S9 Hazard ratios (per SD of the DeSurv linear predictor) for overall survival, before and after adjustment for standard clinical covariates and, where annotated, tumor purity, fitted separately in each cohort by complete-case analysis (per-cell n and events shown). The Covariates column lists the clinical variables entered (stage, grade, age, sex, nodal status, or resection margin, as available). After clinical adjustment the DeSurv score remains independently prognostic in Puleo, PACA-AU array and both training cohorts, and is attenuated to non-significance in PACA-AU RNA-seq, Dijk and Moffitt, the three smallest complete-case sets. Where tumor-purity annotation was available, purity adjustment left the estimate essentially unchanged wherever it could be measured on the full cohort (PACA-AU array 1.44 versus 1.45, PACA-AU RNA-seq 1.66 versus 1.65, TCGA 3.50 versus 3.60; unadjusted versus purity-adjusted), providing no evidence that the association was explained by measured tumor purity. The smaller Dijk and Moffitt cohorts are individually underpowered, and their estimates are correspondingly unstable: Moffitt is non-significant both unadjusted and after adjustment, and its purity-adjusted model rests on only $n = 29$ of its 123 patients. Training-cohort (TCGA, CPTAC) hazard ratios are in-sample. 47

S10 External validation C-index by method and validation dataset. DeSurv at the one-standard-error-selected rank ($k = 3$) is compared with standard NMF at matched rank ($k = 3$) and at elbow-selected rank ($k = 5$), with the matched-budget unsupervised control ($k = 3$, $\alpha = 0$) and with the rank-optimized unsupervised control ($k = 7$, $\alpha = 0$). The unsupervised columns are not all produced by one implementation: the two standard NMF columns are `NMF::nmf`, whereas the two controls are the DeSurv optimizer run at $\alpha = 0$ with the same consensus-initialization procedure as the supervised model. The matched-budget column is the tuning-budget control: it holds rank, optimizer, consensus-initialization procedure and Bayesian-optimization budget fixed while re-optimizing the remaining hyperparameters with $\alpha = 0$. Its external discrimination is no better than that of `NMF::nmf` at the same rank, a comparison made across the two implementations described here. Higher values indicate better discrimination. Every method was scored on the union of its own factors' top-n_{top} gene supports at the supervised model's $n_{\text{top}} = 270$, the same construction as the pooled hazard ratios, main-text Table 1 and Table S11. Scoring the matched-budget control at its own tuned $n_{\text{top}} = 299$ instead changes no per-dataset C-index by more than 0.007; the rank-optimized control was likewise scored at 270 rather than at its own tuned $n_{\text{top}} = 187$. 48

S11 Cohort-specific external-validation estimates for the frozen DeSurv linear predictor, on the de-duplicated patient-level table (570 unique patients, 388 deaths; PACA-AU represented once, RNA-seq retained). Hazard ratios are per standard deviation of the linear predictor computed within that cohort, the convention used for the interaction test, so they are not comparable to the pooled-standard-deviation hazard ratio in main-text Table 1. C-index confidence intervals are percentile intervals from 1000 stratified bootstrap replicates (seed 1) resampling within cohort; the combined C-index is the event-weighted mean of the four cohort values. 49

S12 Top 270 genes for DeSurv factor D1, classical tumor and restCAF-like stroma, ranked by factor specificity score s_{ij}. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension (n_{top}). 50

S13 Top 270 genes for DeSurv factor D2, proCAF-like stroma, ranked by factor specificity score s_{ij}. 51

S14 Top 270 genes for DeSurv factor D3, basal-like tumor, ranked by factor specificity score s_{ij}. 52

Part I

Supplementary Methods

1 Model details

Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix with p genes and n subjects. DeSurv approximates $X \approx WH$ with $W \in \mathbb{R}_{\geq 0}^{p \times k}$ and $H \in \mathbb{R}_{\geq 0}^{k \times n}$, and links the shared program matrix W to survival via an elastic-net–penalized Cox model. Supervision is placed on the program matrix W, the basis that transfers to new cohorts, whereas the sample loadings H serve only to reconstruct the training expression matrix; new samples are therefore scored by projection onto the learned programs, $Z = W^\top X$, rather than by re-estimating loadings.

Let $y \in \mathbb{R}_{>0}^n$ be the observed survival times, $\delta \in \{0,1\}^n$ the event indicators, and $Z = W^\top X \in \mathbb{R}^{k \times n}$ the sample-wise projected scores. We write Z_i for the ith column of Z, and $n_{\text{event}} = \sum_{i=1}^n \delta_i$. For convenience we recall the DeSurv loss from Equation 1 in the main text and rewrite it as

$$\begin{aligned}\mathcal{L}(W, H, \beta) = \\ &(1-\alpha)\left(\frac{1}{2\|X\|_F^2}\|X - WH\|_F^2 + \frac{\lambda_H}{2nk}\|H\|_F^2\right) \\ &-\alpha\left(\frac{2}{n_{event}}\ell(W,\beta) - \lambda\left\{\xi\|b\|_1 + \tfrac{1-\xi}{2}\|b\|_2^2\right\}\right), \qquad b = D_Z\,\beta,\end{aligned}$$

where $\|\cdot\|_F$ is the Frobenius norm, $\beta \in \mathbb{R}^k$ are the Cox coefficients on the raw projected scores, and $D_Z = \operatorname{diag}\big(\operatorname{sd}(Z_{1\cdot}), \ldots, \operatorname{sd}(Z_{k\cdot})\big)$ is the diagonal matrix of the sample-wise standard deviations of the k rows of $Z = W^\top X$ evaluated at the current W. The elastic-net penalty therefore acts on the coefficients of the standardized scores, $b = D_Z\beta$. The inverse map $\beta = D_Z^{-1}b$ is defined only on the retained, nondegenerate score rows, those whose sample standard deviation exceeds the numerical floor stated in Update for β below, and a row falling below that floor is dropped with its coefficient set to zero. This is the objective the implementation evaluates at every acceptance test and records at every outer iteration, with $\operatorname{sd}(Z)$ recomputed from each accepted W (Update for W below), so D_Z changes across iterations rather than being fixed once. The reported λ and ξ are the nominal values passed to the optimizer and are to be read together with this standardization convention. The Cox log-partial likelihood $\ell(W, \beta)$ is

$$\ell(W,\beta) = \sum_{i=1}^n \delta_i \left[Z_i^\top \beta - \log\left(\sum_{j: y_j \geq y_i} \exp(Z_j^\top \beta)\right)\right]. \tag{S1}$$

The inner sum runs over the risk set $\{j : y_j \geq y_i\}$ of subjects still at risk at the observed time y_i; tied event times are handled by the Breslow approximation in the evaluated log-partial likelihood; the derivatives used by the W proposal and by DeSurv's own Coxnet-style coordinate-descent routine for the β update (below) accumulate tied events in data order, so within a tie group they omit the later tied events for all but the last subject. Hyperparameters satisfy $\alpha \in [0,1)$, $\lambda_H \geq 0$, $\lambda \geq 0$, and $\xi \in [0,1]$. The constants $1/(2\|X\|_F^2)$, $\lambda_H/(2nk)$ and $2/n_{\text{event}}$ normalize the component losses, the reconstruction, loadings-penalty and survival terms in that order, before α sets their relative contribution. The elastic-net mixing parameter ξ is the argument `nu` in the DeSurv package. The implementation also carries an optional ridge penalty on W, $\lambda_W\|W\|_F^2/(2pk)$, held at $\lambda_W = 0$ in every reported fit and pinned to the same value throughout the Bayesian-optimization search; it is omitted from the display above.

2 Optimization Algorithm

2.1 Block coordinate descent scheme

Optimization proceeds by block coordinate descent (BCD; Algorithm S1). Each outer iteration updates H, proposes an update to W with the accompanying normalization and coefficient rescaling, and then updates β. The outer loop stops when the *cosine change* of both W and H falls below the tolerance *tol*. For a matrix A we define the cosine change between successive iterates as one minus the cosine similarity of the vectorized iterates,

$$1 - \cos\big(\operatorname{vec} A^{(t)},\, \operatorname{vec} A^{(t+1)}\big) = 1 - \frac{\langle \operatorname{vec} A^{(t)},\, \operatorname{vec} A^{(t+1)} \rangle}{\|A^{(t)}\|_F \, \|A^{(t+1)}\|_F},$$

clipped to $[0, 2]$, so the criterion measures the change in direction of the whole iterate and is invariant to a common rescaling of it. Both quantities must fall below *tol* in the same iteration. A rejected W step (Algorithm S2) leaves W unchanged with cosine change zero, so the rule can also be met by H converging with W fixed, which is how the reported consensus fit terminated. The objective value is recorded once per outer iteration, after the β update, and that recorded series ends with the last outer iteration and does not include the post-loop β refit, whose objective is stored separately.

2.2 Update for H

Conditional on W, the loss $\mathcal{L}(W, H, \beta)$ reduces to a convex quadratic function of H. We adopt the standard NMF multiplicative update with ℓ_2 penalty:

$$H \leftarrow \max\left(H \odot \frac{W^\top X}{W^\top W H + \dfrac{\|X\|_F^2}{nk}\, \lambda_H H + \varepsilon_H}, \varepsilon_H \right), \tag{S2}$$

where $\odot$ denotes elementwise multiplication, all divisions are elementwise, and $\varepsilon_H > 0$ is a small floor to prevent entries from becoming exactly zero. This update is equivalent to a majorization–minimization step and, apart from the ε_H floor, does not increase the reconstruction term (with its λ_H penalty) conditional on W[1,2]. The ε_H-floor ensures that iterates remain in the interior of the nonnegative orthant, which keeps the multiplicative H-update numerically well-defined.

2.3 Update for W

For W, we construct a hybrid multiplicative update that balances the contributions of the NMF and Cox gradients. Let

$$\nabla_W \mathcal{L}_{\mathrm{NMF}}(W, H) = \frac{1}{\|X\|_F^2}(WH - X)H^\top,$$

and let

$$G_\ell(W, \beta) = \frac{2}{n_{\mathrm{event}}} \nabla_W \ell(W, \beta)$$

denote the scaled log-partial-likelihood gradient, where $\nabla_W \ell(W, \beta)$ is the gradient of the Cox log-partial likelihood with respect to W. G_ℓ is not the gradient of the penalized survival component $\mathcal{L}_{\mathrm{Cox}}$ of main-text Equation 1, whose penalty depends on W through D_Z. A closed-form expression for $\nabla_W \ell$ is

$$\nabla_W \ell(W, \beta) = \sum_{i=1}^{n} \delta_i \left(x_i - \frac{\sum_{j: y_j \geq y_i} x_j \exp(x_j^\top W \beta)}{\sum_{j: y_j \geq y_i} \exp(x_j^\top W \beta)} \right) \beta^\top, \tag{S3}$$

where x_i denotes the ith column of X.

Algorithm S1 DeSurv block coordinate descent

Input: $X \in \mathbb{R}_{\geq 0}^{p \times n}$, survival times $y \in \mathbb{R}_{>0}^{n}$, event indicators $\delta \in \{0,1\}^n$, tolerance tol, maximum iterations $maxit$, supervision parameter α, rank k, penalties λ_H, λ, and ξ

Output: Fitted DeSurv parameters (W, H, β)

Initialize $W^{(0)}, H^{(0)}$ with nonnegative entries (single-run fits: Uniform$(0, \max X)$; final model: consensus $W^{(0)}$ with NNLS $H^{(0)}$, Section 5.1), then scale the columns of $W^{(0)}$ to unit norm and the rows of $H^{(0)}$ inversely
Initialize $\beta^{(0)} = \mathbf{0}$ (the value used for the final model)
$eps_W = \infty$, $eps_H = \infty$, $t = 0$
while ($eps_W > tol$ **or** $eps_H > tol$) **and** $t < maxit$ **do**
    (H-update)
        Update $H^{(t+1)}$ from $H^{(t)}$ using the multiplicative rule in Eq. S2,
        holding $W^{(t)}$ and $\beta^{(t)}$ fixed.
    (W-update)
        Update $W^{(t+1)}$ from $W^{(t)}$ using the hybrid multiplicative rule in Eq. S6,
        holding $H^{(t+1)}$ and $\beta^{(t)}$ fixed, and perform backtracking to ensure
        $\mathcal{L}$ does not increase.
    (β-update)
        Update $b^{(t+1)}$ by cyclic coordinate descent on the centered and
        standardized scores (Eq. S7), initialized from the
        current raw coefficients (used as starting b without conversion by D_Z), holding
        $W^{(t+1)}$ and $H^{(t+1)}$ fixed, stopping when the largest scaled
        coefficient change falls below 10^{-6} or after 500 passes, and
        accepting the proposal only if it does not increase the penalized
        survival loss (otherwise a damped step toward it, or the previous
        b retained); set $\beta^{(t+1)} = D_Z^{-1} b^{(t+1)}$ on the retained rows.
    (Convergence)
        $eps_W = 1 - \cos\big(\operatorname{vec} W^{(t)}, \operatorname{vec} W^{(t+1)}\big)$,
        $eps_H = 1 - \cos\big(\operatorname{vec} H^{(t)}, \operatorname{vec} H^{(t+1)}\big)$
        Record $\mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)})$ in the objective series
    $t = t + 1$
end while
Refit β on the final W by the same routine, giving β_{refit} (this is the only β estimate when $\alpha = 0$)
The objective after the refit is stored separately and is not part of the recorded series
return $\hat{W} = W^{(t)}, \hat{H} = H^{(t)}, \hat{\beta} = \beta_{\text{refit}}$

We define a gradient-scaling heuristic

$$\zeta^{(t)} = \frac{\left\|(1-\alpha)\nabla_W \mathcal{L}_{\mathrm{NMF}}(W^{(t)}, H^{(t+1)})\right\|_F + \varepsilon_\zeta}{\left\|\alpha\, G_\ell(W^{(t)}, \beta^{(t)})\right\|_F + \varepsilon_\zeta}, \tag{S4}$$

with a numerical stabilizer $\varepsilon_\zeta = 10^{-12}$ in each norm, and clip $\zeta^{(t)}$ to avoid extreme values: $\zeta^{(t)} \leftarrow \min(\zeta^{(t)}, \zeta_{\max})$ with $\zeta_{\max} = 10^6$ in all experiments; below we write ζ for $\zeta^{(t)}$ where the iteration index is not needed. The proposal rescales the weighted survival gradient by this norm-based factor, leaving the reconstruction terms unscaled, subject to its numerical stabilizer ε_ζ and the cap $\zeta_{\max}$, before the additional factor α below controls the relative Cox contribution. Because the Cox gradient entering ζ is already weighted by α, the additional explicit factor of α in the W update below makes the scaled Cox contribution approximately α times the weighted reconstruction-gradient norm when the cap is inactive, up to the numerical stabilizer; the two factors of α in that update are therefore not a duplication.

In the reported consensus fit, ζ decreased from 0.71 after the first iteration to 0.059 at convergence, and the scaled Cox contribution was then 1.6% of the reconstruction term in the numerator. The cap was reached only at initialization, where $\beta = 0$ makes the Cox gradient itself zero, so the cap contributed no update; across the 100 single-run endpoints ζ ranged from 0.003 to 0.031 and was never capped (the ζ-trace script, Table S5, reconstructs the production optimization path from the cached seed fits and checks it against the cached final fit to a declared tolerance).

The multiplicative factor for W is then

$$R^{(t)} = \frac{\frac{(1-\alpha)}{\|X\|_F^2} X H^{(t+1)\top} + \alpha\, \zeta^{(t)}\, \frac{2\alpha}{n_{\mathrm{event}}}\, \nabla_W \ell(W^{(t)}, \beta^{(t)})}{\frac{(1-\alpha)}{\|X\|_F^2} W^{(t)} H^{(t+1)} H^{(t+1)\top} + \varepsilon_W}, \tag{S5}$$

$R^{(t)}$ is clamped elementwise to $[0.2, 5]$ before use, which bounds the change of any entry of W in a single step, and the proposed update is

$$W^{(t+1)} = \max\Big(W^{(t)} \odot R^{(t)}, \varepsilon_W\Big), \tag{S6}$$

with a small floor $\varepsilon_W > 0$. When $\alpha = 0$, $R^{(t)}$ reduces to the factor of the standard multiplicative update for NMF[1]; for $\alpha > 0$, the Cox gradient perturbs the update in a direction that increases the log-partial likelihood ℓ to first order.

The proposal uses the reconstruction and Cox log-partial-likelihood derivatives only. It does not differentiate the W-dependent penalty on the standardized-score coefficients $b = D_Z\beta$, so it is not the gradient of the full objective $\mathcal{L}$, and candidate acceptance is instead evaluated using the full objective. Because the survival derivative enters the proposal with weight $\alpha^2\zeta$ rather than α, a fixed point of the proposal ($R^{(t)} = 1$ on the entries above ε_W) is not in general a stationary point of $\mathcal{L}$, even setting the D_Z penalty aside.

Because the combined multiplicative step does not, by itself, guarantee a nonincrease in the full loss $\mathcal{L}$, we embed it in a backtracking line search.

Algorithm S2 W update with backtracking

Input: W at the previous iteration t, $W^{(t)}$; the multiplicative update term at the current iteration $R^{(t)}$, the max number of backtracking updates max_bt, the initial step $s_0 \in (0, 1]$, the backtracking factor $\gamma \in (0, 1)$

Output: $W^{(t+1)}$

$s = s_0$
$m = 0$
$flag_accept = FALSE$
while $m \leq max_bt$ **do**
  $W^{(t+1)}_{cand} = \max\big(W^{(t)} \odot [(1 - s) + sR^{(t)}],\ \varepsilon_W\big)$
  (Column normalization)
    Compute $D = \mathrm{diag}\big(\|W^{(t+1)}_{(:,1)cand}\|_2, \ldots, \|W^{(t+1)}_{(:,k)cand}\|_2\big)$
    and set $W^{(t+1)}_{cand} \leftarrow W^{(t+1)}_{cand} D^{-1}$, $H^{(t+1)}_{cand} \leftarrow DH^{(t+1)}$, and $\beta^{(t)}_{cand} \leftarrow D\beta^{(t)}$
  if $\mathcal{L}(W^{(t+1)}_{cand}, H^{(t+1)}_{cand}, \beta^{(t)}_{cand}) \leq \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)})$ **then**
    $W^{(t+1)} = W^{(t+1)}_{cand}$
    $H^{(t+1)} = H^{(t+1)}_{cand}$
    $\beta^{(t)} = \beta^{(t)}_{cand}$
    $flag_accept = TRUE$
    break
  end if
  $s = s\gamma$
  $m = m + 1$
end while
if $flag_accept = FALSE$ **then**
  $W^{(t+1)} = W^{(t)}$
end if

In all reported fits $s_0 = 0.5$, $\gamma = 0.5$ and $max_bt = 10$ (the values passed by the package's R wrapper to the optimizer). The column normalization inside the line search is described in Normalization of W below; it is applied to each candidate before the objective is compared, so the accepted update is the normalized one. The score standard deviations D_Z that enter the penalty (Model details) are recomputed from each candidate, as $\mathrm{sd}(Z_{j\cdot})$ with $Z = W^{\top}_{cand}X$, and are carried forward only when that candidate is accepted; a rejected candidate leaves W, H, β and D_Z at their previous values.

2.4 Update for β

Conditional on (W, H), the loss in β reduces to a convex elastic-net–penalized Cox problem in the standardized-score coefficients $b = D_Z\beta$; the common factor $\alpha > 0$ multiplies both the log-partial-likelihood and the penalty in the joint loss (main-text Equation 1) and is dropped here, as it does not affect the minimizer,

$$\min_{b \in \mathbb{R}^k} \left\{ -\frac{2}{n_{\text{event}}}\, \ell(W, D_Z^{-1}b) + \lambda \left(\xi \|b\|_1 + \frac{1 - \xi}{2} \|b\|_2^2 \right) \right\},$$

with D_Z held at the value implied by the current W. The coefficients reported throughout the manuscript are the raw-score coefficients $\beta = D_Z^{-1}b$ on the retained rows. We update b by cyclic coordinate descent following[3]. Let $\tilde{z}_i = D_Z^{-1}(Z_i - \bar{Z})$ denote the centered and standardized score vector of sample i, with Z_i the ith column of the raw scores Z and $\bar{Z}$ the vector of row means, and write $\tilde{\eta}_i = \tilde{z}_i^{\top} b$, which equals $Z_i^{\top}\beta$ up to an additive constant common to all samples. The update for coordinate r has the closed form

$$\hat{b}_r = \frac{S\left(\frac{1}{n_{\text{event}}} \sum_{i=1}^{n} w(\tilde{\eta})_i \tilde{z}_{i,r} \left[v(\tilde{\eta})_i - \sum_{j \neq r} \tilde{z}_{i,j} b_j \right], \lambda\xi \right)}{\frac{1}{n_{\text{event}}} \sum_{i=1}^{n} w(\tilde{\eta})_i \tilde{z}_{i,r}^2 + \lambda(1 - \xi)}, \tag{S7}$$

where $S(a, \tau) = \text{sign}(a)\max(|a| - \tau, 0)$ is the soft-thresholding operator, $w(\tilde{\eta})_i$ and $v(\tilde{\eta})_i$ are the standard local quadratic-approximation weight and working response for sample i in the Cox log-partial likelihood[3], and $\tilde{z}_{i,r}$ is the rth entry of $\tilde{z}_i$. After the sweeps, the raw-score coefficients are recovered as $\beta = D_Z^{-1} b$ on the retained rows.

The sweeps operate on centered and standardized scores, the parameterization in which $\mathcal{L}$ is defined. The k rows of Z are centered and scaled to unit standard deviation across samples, rows whose standard deviation falls below 10^{-12} are dropped and assigned coefficient zero, the sweeps estimate b, and the estimates are divided back by the same standard deviations to give the reported $\beta = D_Z^{-1} b$ on the retained rows. Each update is initialized from the current raw coefficients, taken after the W-normalization step and used as starting values of b without conversion by D_Z. Centering the scores changes the linear predictor by a common additive constant, which cancels from the Cox partial likelihood. Given that parameterization, the implemented update departs from an exact conditional minimization of $\mathcal{L}$ in two ways, which we state explicitly because Equation 1 of the main text, reproduced as $\mathcal{L}$ above, is the objective the algorithm evaluates and records. First, the quadratic approximation entering Eq. S7 carries likelihood weight $1/n_{\text{event}}$, whereas the log-partial likelihood in the objective carries weight $2/n_{\text{event}}$; relative to that likelihood weight the penalty seen by the coordinate sweeps is therefore effectively 2λ, so the sweeps target a more strongly penalized subproblem than the conditional restriction of $\mathcal{L}$. Second, the sweeps are iterated until the largest scaled coefficient change falls below 10^{-6} or 500 passes have been taken, and the resulting proposal is accepted only if it does not increase the penalized survival loss; otherwise a damped step toward it is taken (backtracking factor 0.9, at most 30 steps), or the previous β is retained. That acceptance safeguard makes the β update non-increasing in the evaluated objective, and it does not establish convergence to a conditional or joint optimum. The β block is therefore an approximate, monotone conditional update rather than an exact conditional minimizer of $\mathcal{L}$. The acceptance safeguards prevent accepted W and β updates from increasing the evaluated objective; they do not establish convergence to an optimum. The reported fits are exactly those produced by this procedure. After the outer loop terminates, β is refit once on the final W by the same routine under the same tolerance and pass limit (Algorithm S1); this refit is the reported $\hat{\beta}$, and it is the only β estimate when $\alpha = 0$. The objective after that refit is stored separately and is not part of the recorded outer-loop series.

2.5 Normalization of W

Within each candidate W update, we normalize the columns of W as $W \leftarrow WD^{-1}$, and adjust H and β as

$$H \leftarrow DH, \qquad \beta \leftarrow D\beta,$$

where D is the diagonal matrix containing the column Euclidean norms of the candidate W. Because D has positive diagonal entries, these compensating rescalings preserve the reconstruction WH, the collapsed gene-weight vector $W\beta$ and hence the linear predictor $X^\top W\beta$, and also the standardized-score coefficients $b = D_Z\beta$, because D_Z is rescaled by D^{-1} exactly as β is rescaled by D. With $\lambda_H = \lambda_W = 0$, as in every reported fit, normalization alone therefore leaves the displayed objective unchanged. Each normalized candidate is nonetheless evaluated under the full objective within the backtracking procedure for the W update rather than assumed loss-invariant. Normalization prevents degeneracy in the scale-nonidentifiable factorization and keeps columns of W comparable in magnitude and interpretable as gene programs.

3 Supervision on W versus H

Classical nonnegative matrix factorization (NMF) is non-identifiable: for any invertible matrix R with $WR \geq 0$ and $R^{-1}H \geq 0$, the factorization (W, H) can be transformed to $(WR, R^{-1}H)$ without changing the reconstruction error, yielding multiple valid solutions[4–6]. Although normalizing columns of W or rows of H resolves the trivial scaling ambiguity, nonnegative mixing transformations persist, so factorizations started from different initializations need not agree[7–9]. This applies to DeSurv as well: individual restarts of

the reported model are dispersed, and the final fit is obtained by a consensus procedure over many restarts (Stability of the recovered gene programs).

In DeSurv, survival supervision enters through the projection $Z = W^\top X$ in the partial log-likelihood, so that both component losses have a gradient with respect to the programs. The update for W combines the weighted reconstruction gradient with the scaled log-partial-likelihood gradient G_ℓ through the rescaled multiplicative construction of Equations S4 to S6, so the gene-program definitions are shaped directly by their association with survival. That construction is a proposal direction and not the gradient of the full objective, which depends on W additionally through the penalty on $b = D_Z\beta$ (Update for W).

Related supervised-NMF methods instead place the survival term on the sample side. CoxNMF maximizes a Cox partial likelihood whose covariates are the columns of H[10], and SurvNMF jointly learns the factorization and a Cox model on the sample-side latent matrix, which its own transposed notation writes as W[11]. Because the risk score under either formulation is a function of the sample loadings rather than of the gene-program basis, a new sample needs loadings of its own. SurvNMF documents such a step, obtaining them for held-out samples by a constrained least-squares fit; that step holds the gene basis fixed and needs neither relearning of the basis nor validation survival outcomes, so it is an additional fitting step at transfer rather than a leak. CoxNMF's published description does not specify a new-sample scoring procedure.

With supervision applied to the sample loadings H, the Cox term has no direct gradient with respect to the gene-program matrix W; survival can nevertheless influence W indirectly through the alternating reconstruction updates involving H. The distinction concerns where supervision enters the objective, not a guarantee that sample-side supervision leaves the basis unchanged; related considerations arise in supervised topic models[12].

Supervising W also provides a practical advantage for **portability**: because W defines gene-level programs, new samples can be scored directly by the projection $Z_{\text{new}} = W^\top X_{\text{new}}$. Supervising H would not yield such a mapping: obtaining sample loadings for external datasets would require an additional sample-specific estimation step, a nonnegative least-squares (NNLS) problem for each new sample. SurvNMF uses constrained least squares for held-out samples; CoxNMF does not describe new-sample scoring in the cited source (Supplementary Table S1). Estimating loadings with a fixed basis does not require relearning the gene programs or using validation survival outcomes.

Supervising through W is therefore intended to shape the gene programs toward prognostic directions and to yield signatures that can be scored unchanged in new cohorts; whether they do so is an empirical question addressed in Part II.

Supplementary Table S1 places DeSurv beside the unsupervised factorization it extends, the two published survival-supervised factorizations, and the two conventional supervised predictors used as comparators, by where survival enters each method, what object is learned, and what must happen when a new sample arrives. It records architecture only; it does not rank the methods.

4 Cross-validation procedure

Algorithm S3 describes the cross validation procedure with F folds and R initializations. We define the C-index using comparable pairs $\mathcal{P} = \{(i,j) : y_i < y_j,\ \delta_i = 1\}$ and linear predictors $\hat{\eta}_i$:

$$\hat{c} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left[\mathbb{1}(\hat{\eta}_i > \hat{\eta}_j) + \tfrac{1}{2}\mathbb{1}(\hat{\eta}_i = \hat{\eta}_j) \right]. \tag{S8}$$

Algorithm S3 Cross-validation for DeSurv pipeline

Input:

Number of folds F

Number of initializations R

Observed data (X, y, δ)

Hyperparameters k, α, λ, ξ, λ_H, and number of top genes per factor n_{top}

Output: The cross-validated C-index

Divide subjects into F folds.
for $f = 1$ to F **do**
  Split data into training and validation $(X_{(-f)}, y_{(-f)}, \delta_{(-f)})$, and $(X_{(f)}, y_{(f)}, \delta_{(f)})$ sets
  for $r = 1$ to R **do**
    set.seed(r)
    Apply Algorithm S1 with inputs $(X_{(-f)}, y_{(-f)}, \delta_{(-f)})$, and $(k, \alpha, \lambda, \xi, \lambda_H)$
    Obtain $\hat{W}$ and $\hat{\beta}$ as output from Algorithm S1
    Restrict $\hat{W}$ to the union of the k factors' top-n_{top} gene sets (by factor-specificity score) and compute the estimated linear predictor on this union support: $\hat{\eta} = X_{(f)}^T \hat{W} \hat{\beta}$
    Compute C-index, denoted $\hat{c}_{(f)r}$, according to Equation S8
  end for
  end loop over r
  Compute average C-index across initializations: $\hat{c}_{(f)} = \frac{1}{R} \sum_{r=1}^{R} \hat{c}_{(f)r}$
end for
end loop over f
Compute final cross-validated C-index: $\hat{c} = \frac{1}{F} \sum_{f=1}^{F} \hat{c}_{(f)}$
return Final estimate $\hat{c}$

5 Bayesian Optimization

We treat hyperparameter tuning as a black-box optimization problem over a bounded domain. Let ψ collect the tuned parameters (at minimum $\psi = (k, \alpha, \lambda, \xi)$, and optionally discrete choices such as the signature size n_{top}). For a given ψ we run the cross-validation procedure in Algorithm S3, averaging the C-index across folds and random initializations to form the objective

$$\hat{c}(\psi) = \frac{1}{F} \sum_{f=1}^{F} \hat{c}_{(f)}(\psi).$$

Because $\hat{c}(\psi)$ is expensive to evaluate and non-differentiable with respect to ψ, we use Bayesian optimization (BO) to adaptively select candidate configurations.

We place a Gaussian-process (GP) prior on the response surface $f(\psi) \sim \mathcal{GP}(0, k(\psi, \psi'))$ with a Matérn kernel and automatic relevance determination length-scales. Continuous parameters are rescaled to $[0, 1]$ and penalties are optimized on a log scale; integer-valued coordinates (e.g., k or n_{top}) are proposed in the continuous space and rounded inside the evaluator. Following an initial batch of evaluations, the GP is refit each time a new observation is added and used to guide acquisition.

At iteration t, let $\mu_t(\psi)$ and $\sigma_t(\psi)$ denote the GP posterior mean and standard deviation, and let $f_\star$ be the best observed score. We select new candidates by maximizing expected improvement (EI),

$$\text{EI}(\psi) = (\mu_t(\psi) - f_\star - \kappa)\, \Phi(Z(\psi)) + \sigma_t(\psi) \phi(Z(\psi)), \quad Z(\psi) = \frac{\mu_t(\psi) - f_\star - \kappa}{\sigma_t(\psi)},$$

where Φ and ϕ are the standard normal cdf/pdf and $\kappa \geq 0$ controls the exploration margin. EI is evaluated over a candidate pool sampled from the bounded search region, and the maximizer is evaluated next. The loop continues until the iteration budget is exhausted or the improvement plateaus.

For the PDAC application, the objective was the mean 5-fold cross-validated Harrell C-index (Algorithm S3), with five-fold cross-validation constructed separately within cohort and stratified by event status under a fixed seed (123); each configuration was evaluated from 30 random initializations, as in the definition above. Cross-validation fits used a stopping tolerance of 10^{-5} on the cosine change of W and of H, looser than the 10^{-7} used for the restart and final fits, under the same cap of 4,000 outer iterations. The bounded search domain was $k \in [2, 12]$, $\alpha \in [0, 1]$ (the proposal bounds; the model domain is $\alpha \in [0, 1)$, package validation clips proposals at $\alpha \geq 1$, and the largest evaluated value was 0.986), $\lambda \in [10^{-3}, 10^{3}]$ (optimized on a $\log_{10}$ scale), $\xi \in [0, 1]$, and $n_{\text{top}} \in [50, 300]$, with the loadings ridge penalty fixed at $\lambda_H = 0$. Bayesian optimization used an initial design of 50 evaluations followed by 100 GP-guided iterations (150 total evaluations), maximizing expected improvement over a candidate pool of 4,000 points sampled from the bounded domain at each step with an exploration margin $\kappa = 0.01$. The unsupervised comparator fixed $\alpha = 0$ and tuned the remaining coordinates identically.

For model selection, we take the highest observed $\hat{c}(\psi)$ as the best observed configuration; because the search is a finite, stochastic exploration of a black-box surface, this is not a claim of a global optimum. When fold-level standard errors are available, we apply a one-standard-error rule for k: for each rank we retain the configuration with the highest observed mean cross-validated C-index, set the threshold one fold-level standard error below the best observed configuration's mean, and select the smallest rank whose retained configuration meets it.

Two details of this rule matter for reading the reported numbers. First, the comparison is between the best configuration observed at each rank: the highest mean cross-validated C-index anywhere in the search was 0.655 at $k = 7$ (fold-level standard error 0.011), and the best configuration observed at $k = 3$ reached 0.646, above the threshold of 0.644. Second, the final model is a hybrid of the two: the rule fixed $k = 3$, while the remaining hyperparameters ($\alpha = 0.334$, $\lambda = 0.349$, $\xi = 0.056$, $n_{\text{top}} = 270$) were retained from the overall best configuration at $k = 7$. The final configuration was therefore not itself a point evaluated in the search. Its cross-validated performance under the identical protocol and the current software build is reported in Table S2: 0.642 (fold-level standard error 0.007), 0.002 below the recomputed one-standard-error threshold of 0.644. Separately, we ran a sensitivity analysis of rank selection using the same training data and the selected non-rank hyperparameters, the fixed-hyperparameter grid of Cross-validated C-index across factorization rank. That grid is an exhaustive rank-by-supervision search over $k \in \{2, 3, \ldots, 12\}$ and $\alpha \in \{0, 0.05, \ldots, 0.95\}$ holding λ, ξ and n_{top} at the selected model's values, and the statements here are restricted throughout to $n_{\text{top}} = 270$. Applying the same one-standard-error rule to it also selects $k = 3$, its best configuration being at $k = 9$ with mean C-index 0.671 and standard error 0.009, and $k = 3$ the smallest rank whose best configuration reaches the resulting threshold 0.662. That corroborates the selected rank without evaluating the final configuration itself, whose nearest evaluated neighbor is the slice of the same grid at the selected rank. At $k = 3$ and $\alpha = 0.35$ the five-fold cross-validated C-index was 0.656 (standard error 0.010), which clears the Bayesian-optimization threshold of 0.644 but not this grid's own one-standard-error threshold of 0.662, and which is a nearby supervision level rather than the selected configuration. The final DeSurv model is refit at the selected parameters using the consensus-based initialization (Consensus initialization for the final model), and the BO-chosen n_{top} is used to truncate each program for downstream analysis.

The parameter n_{top} determines the gene support used for projection and, in the final consensus procedure, the gene sets from each restart that enter the consensus initialization. After factorization, the support is restricted to the union of the k programs' top-n_{top} gene sets, ranked within each program by specificity score s_{ij} (Supplementary Information, Section 8), giving the truncated matrix $\widetilde{W}$; each sample's score for program j is then $\widetilde{W}_j^{\top} x$, the jth column of $\widetilde{W}$ applied to that shared support. Genes outside the union still shape the factorization but do not enter the linear predictor. This union construction is the convention used by the cross-validation objective and by the primary external-validation scoring. BO selects n_{top} jointly with k, α, λ, and ξ by optimizing cross-validated C-index. Each comparator search selects its own n_{top}. The primary factorization comparison used a common number of top genes per program when constructing each model's scoring union, the supervised model's $n_{\text{top}} = 270$ (Supplementary Table S10), with the tuning-selected values reported alongside it; total union size can still differ with rank and overlap among programs. When $\alpha = 0$ (no survival supervision), survival does not enter the factorization objective; it is used for the post hoc Cox fit and for tuning n_{top}, which sets the projection support and, through the consensus initialization, can still

influence the final unsupervised fit indirectly.

Supplementary Fig. S1 illustrates the tuned response surface for the two coordinates of primary interest in the PDAC training cohort: the cross-validated C-index over factorization rank k and supervision strength α. Performance is high across a broad region at intermediate supervision and moderate rank, with a second high region at $k \geq 9$ and $\alpha \approx 0.2$–0.3, and degrades toward the unsupervised corner ($\alpha = 0$), consistent with a benefit of survival supervision; the second region is consistent with the flat rank dependence of the supervised curve in Section 24. This is an empirical model-selection sensitivity surface, not a simulation result.

5.1 Consensus initialization for the final model

The final DeSurv model at the selected hyperparameters was initialized from a consensus of multiple independent runs to reduce sensitivity to random starts. We first fit $R = 100$ DeSurv models to the training data from independent random nonnegative initializations at the selected $(k, \alpha, \lambda, \xi, n_{\text{top}})$ (seeds 1–100). For each run, the top n_{top} genes of every learned program were extracted by factor-specificity score, and a gene-by-gene consensus matrix was accumulated across runs, recording how often each pair of genes appeared together among the top genes of the same program.

Genes appearing in the top sets of fewer than $\lceil 0.3R \rceil$ of the R runs were treated as inactive and excluded from clustering. Because the consensus is constructed from gene-by-gene co-occurrence across restarts, it is invariant to factor-label permutations and does not require explicit alignment of factor labels across runs. The consensus matrix over the remaining genes was partitioned into k groups by agglomerative hierarchical clustering (average linkage), and each cluster defined one column of the initial gene-program matrix W_0; inactive genes and otherwise-empty entries were set to a small positive floor ($\sqrt{\epsilon_{\text{mach}}}$) to preserve strict positivity. The initial loadings H_0 were obtained by nonnegative least squares given W_0 and the training expression matrix, and the initial Cox coefficients β_0 were set to zero. This consensus seed used only training-cohort data and was applied solely to initialize the single final refit after hyperparameter selection; it used no validation-cohort data and therefore introduced no information leakage.

6 PDAC Datasets

We analyzed PDAC cohorts from TCGA-PAAD and CPTAC-PDAC as training data[13,14]. External validation used independent cohorts from Dijk, Moffitt, PACA-AU (array and RNA-seq), and Puleo[15–18]. Training models were fit on the combined TCGA and CPTAC expression matrices after harmonizing gene identifiers; validation datasets were processed with the same gene set and transformations before projection, scoring and downstream validation analyses.

Dataset-specific inclusion and preprocessing steps were as follows. For TCGA-PAAD, we excluded samples lacking tumor grade and mapped features to gene symbols. For CPTAC, we retained samples annotated as PDAC by histology. For Dijk, all 90 samples passed quality-control filters and were retained. For Moffitt, only primary tumor specimens were kept. For PACA-AU array and RNA-seq, we retained primary PDAC tumors and collapsed duplicated gene symbols by median. Puleo required no additional dataset-specific filtering beyond survival QC.

Across cohorts, samples without valid survival outcomes (nonpositive time, missing time, or missing event indicator) were removed. Expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort (TCGA-PAAD and CPTAC) separately, genes were ranked in descending order of mean expression and, independently, of variance; the 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection of the two cohorts' gene sets yielded 1,970 genes used for model training. Validation datasets were restricted to this same 1,970-gene set.

A within-sample rank transformation was then applied to every dataset to mitigate platform and scale effects. Specifically, for each sample the expression values across the 1,970 retained genes were replaced by

their ascending ranks (ties resolved by average ranks), so that every sample carries the same nonnegative rank distribution over the shared gene set; ranks are inherently nonnegative and therefore preserve NMF compatibility. Validation cohorts were rank-transformed by exactly the same per-sample procedure before projection, with no use of validation outcomes.

Table S3 summarizes the cohorts, platforms, sample sizes after quality control, and patient-level metadata used in this study. Overall survival time and event indicators were available for all cohorts. PurIST[19] and DeCAF[20] molecular classifications, applied to bulk expression data as described in those references, were used as covariates in the covariate-adjusted external validation analysis. Gene expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study[20]; training data were obtained directly from GDC.

7 Survival Analysis

Survival analyses used overall survival time with event indicators as provided by each cohort. For fitted DeSurv models, sample-level program scores are the raw projections $Z = \widetilde{W}^\top X$ onto the frozen support, and the linear predictor is $\hat{\eta} = Z^\top \hat{\beta}$ with the trained coefficients; no re-estimation and no standardization enters the score itself. The frozen support is the union of the k programs' top-n_{top} gene sets, and $\widetilde{W}$ is $\hat{W}$ restricted to those rows, so every column keeps its own weights on the whole union and a gene entering the union through one program still contributes to the other programs' scores with its own weight. This union construction defines the primary external-validation scores, including the headline pooled hazard ratio, the leave-one-dataset-out estimates, the classifier-adjusted risk-score analyses, the per-factor forest estimates and the cohort-specific estimates of Table S11. Other analyses use explicitly identified alternative constructions, named where they occur. Full-support scores over all analyzed genes are used for the training-data contribution analysis (Reconstruction and Survival Contribution per Factor) and for the frozen-axis comparator analysis whose subsampling stability is reported in Stability of the recovered gene programs. Per-program masks, each column zeroed outside its own top-n_{top} set, are used for the tuned supervised-comparator analysis (Robustness to tuned supervised comparators and compartment attribution) and for descriptive checks such as Cohort handling in the training objective below. Both of those supervised-comparator constructions, the one on the full shared support and the one on per-program masks, additionally refit unpenalized Cox coefficients on the training scores rather than carrying $\hat{\beta}$ forward. Throughout, projection denotes multiplication by this fixed gene-weight matrix; the truncated columns are not renormalized, so the term does not imply an orthogonal geometric projection. The Cox coefficients $\hat{\beta}$ were estimated during model fitting from the full-support scores $W^\top X$ over all analyzed genes; for cross-validation and external scoring they were carried forward unchanged and applied to scores from the truncated basis $\widetilde{W}$, and were not refit after truncation (Algorithm S3). The elastic-net penalty set the trained coefficient for D2 to exactly zero, so $\hat{\eta}$ is a weighted sum of the D1 and D3 scores. Prognostic performance was summarized using the concordance index as defined above (Cross-validation procedure). When multiple datasets were pooled, Cox models were stratified by dataset to allow cohort-specific baseline hazards.

For a fixed fitted predictor and scoring support, the manuscript uses the following reporting scalings. Every linear predictor built on the union support is the same linear functional of expression up to a positive constant, and the map from scaling to reported quantity is as follows. (i) During cross-validated tuning the collapsed weight vector $\widetilde{W}\hat{\beta}$ is scaled to unit ℓ_2 norm (main text, Methods); this changes $\hat{\eta}$ by a positive constant only. (ii) Primary risk-score hazard ratios, namely the headline pooled estimate, the covariate-adjusted estimates of main-text Table 1 and the leave-one-dataset-out estimates, divide $\hat{\eta}$ by its standard deviation over the pooled, de-duplicated validation set, so they are per pooled standard deviation. (iii) Two further analyses standardize to zero mean and unit standard deviation within groups, and the grouping differs between them. The per-factor forest estimates (main-text Fig. 3a) and the D1 classifier-adjustment analysis (Variance partition of the D1 score) standardize within each of the five platform datasets, whereas the between-cohort slope analysis, its interaction test and the cohort-specific hazard ratios of Table S11 standardize within each of the four patient cohorts, so these are per within-dataset and per within-cohort standard deviation respectively. Hazard ratios under (ii) and (iii) are not interchangeable. Within a given cohort or dataset,

centering and positive rescaling of a fixed score leave its C-index unchanged; this does not equate different predictors and does not preserve pooled rankings across groups standardized differently. Analyses that use a different gene support, listed above, are not scalings of this functional and are reported as separate quantities.

Two further conventions apply to the primary combined external C-index and its confidence interval. The combined C-index is the event-weighted mean of the four cohort C-indices rather than a concordance computed over pooled patients, so no concordance pair is formed across cohorts. Its confidence interval is a percentile interval from a stratified nonparametric bootstrap that resamples patients with replacement within cohort ($B = 1{,}000$ replicates, seed 1), so resampling preserves cohort membership (Between-cohort heterogeneity and C-index confidence intervals).

7.1 Independence of discovery and validation

DeSurv is trained with survival supervision only; it never uses PurIST, DeCAF, or any molecular subtype label during factorization or hyperparameter selection. The PurIST and DeCAF subtype calls were fixed before the DeSurv analysis (loaded as precomputed per-cohort annotations) and enter only as covariates in the adjusted Cox models, not as inputs to DeSurv. All rank and hyperparameter selection (including the one-standard-error rule) was performed by cross-validation entirely within the TCGA and CPTAC training data; no external validation cohort was used for training or tuning. Validation-cohort expression enters only at scoring, a fixed-basis projection ($Z = \widetilde{W}^{\top} X_{\text{new}}$) that uses no validation outcome, and validation-cohort survival outcomes were used only for the final, out-of-sample performance evaluation.

7.2 Cohort handling in the training objective

The training Cox objective was fit to the pooled TCGA-PAAD and CPTAC samples without a cohort term; cross-validation folds were constructed within cohort with stratification by event status. Because survival shapes W, a cohort effect associated with both expression and survival could in principle be rewarded by the survival gradient. As a descriptive check for cohort-driven association, computed from the cached training data and final fit and not a refit, the fitted linear predictor remained strongly associated with survival within TCGA-PAAD and CPTAC separately (in-sample HR per SD 3.50 and 3.72; Table S9) and in a model stratified by cohort (HR per SD 3.35, against 3.12 unstratified), while the between-cohort shift in the score was opposite to the between-cohort hazard difference: the predictor averaged 0.73 SD higher in TCGA-PAAD, yet CPTAC had the higher hazard given the predictor (HR 1.90). These in-sample quantities argue against the specific mechanism in which the survival gradient learns cohort membership, but they do not rule out cohort influence on the factorization, whose impact on transportability is tested by external validation.

8 Gene Program Correlation Analysis

To assess the biological identity of learned factors, we evaluated the correspondence between factor-specific gene rankings and established PDAC gene programs. Each column j of the gene program matrix W was first scaled by its column maximum, $W^*_{ij} = W_{ij} / \max_i W_{ij}$, placing all factors on a common $[0, 1]$ scale, and the specificity score for gene i in factor j was defined as

$$s_{ij} = W^*_{ij} - \max_{j' \neq j} W^*_{ij'}.$$

Genes were ranked in descending order of s_{ij}. This ranking defines both the top-n_{top} genes used to score samples (Supplementary Data: factor-specific gene lists) and, for the top 50 genes per factor, the loadings used to quantify factor–program correspondence.

For each factor j and each reference gene program G_k, we computed the Spearman rank correlation between the continuous factor loadings $W_{\cdot j}$ (restricted to the union of the per-factor top-50 gene sets intersected with the reference programs) and a binary indicator vector $\mathbf{1}[\text{gene} \in G_k]$ (main-text Fig. 2a,b; Methods). P

values were derived from the asymptotic Spearman test and adjusted for multiple comparisons using the Benjamini–Hochberg procedure within each factor column independently; as in the main text, these adjusted P values are displayed as descriptive marks, because the gene universe is itself selected by loading. Reference programs were included in the visualization only if at least one factor yielded a Spearman correlation exceeding 0.2.

The reference universe comprises 45 published PDAC gene programs (Table S4): the PurIST basal-like and classical tumor signatures[19]; the DECODER compartment signatures[21]; the Moffitt tumor, stroma and immune programs[16]; the Collisson subtypes[22]; the Bailey subtypes, including ADEX (aberrantly differentiated endocrine exocrine)[17]; the Chan-Seng-Yue subtypes[23]; the Maurer ECM and immune programs[24]; the five Puleo whole-tumor components together with Puleo's two tumor-compartment signatures[18]; the Elyada iCAF and myCAF signatures[25]; the SCISSORS fibroblast and tumor signatures[26]; and the DeCAF proCAF and restCAF programs[20]. Nine source entries were excluded before analysis: six that duplicate a retained list gene for gene (a second copy of the Moffitt activated and normal stroma lists, the SCISSORS CAF-versus-perivascular variants of its apCAF, iCAF and myCAF lists, and a second copy of its perivascular list), the two nine-gene PurISS iCAF and myCAF lists, and Bailey's non-unique gene set. The SCISSORS iCAF and myCAF signatures are displayed under the DeCAF nomenclature, restCAF and proCAF (Stromal resolution: which reference programs are recovered from bulk). Puleo's whole-tumor pure basal-like component and its tumor-compartment basal-like signature are distinct programs and are labeled as such in every figure. Many programs place only a minority of their genes inside the 1,970-gene analysis set; the correlation for a program is computed over the genes it shares with that set, and a program with no such genes cannot be represented.

9 Reconstruction and Survival Contribution per Factor

NMF does not provide an orthogonal variance decomposition; it seeks a low-rank reconstruction of the expression matrix $X \approx WH$. We therefore measure each factor's contribution to reconstruction rather than to variance explained. For a subset of factors $S \subseteq \{1, \dots, k\}$, define the reconstruction value

$$v(S) = \|X\|_F^2 - \Big\| X - \sum_{j \in S} W_{\cdot j} H_{j\cdot} \Big\|_F^2,$$

the reduction in residual error relative to the null (zero-factor) model, where $W_{\cdot j} H_{j\cdot}$ is the rank-one outer-product contribution of factor j (the outer product of column j of W with row j of H). The reconstruction Shapley value[27] of factor j,

$$\phi_j = \sum_{S \subseteq \{1,\dots,k\} \setminus \{j\}} \frac{|S|!\,(k - |S| - 1)!}{k!} \big(v(S \cup \{j\}) - v(S) \big),$$

is its average marginal reconstruction gain over all orderings of the factors.

By the Shapley efficiency axiom, $\sum_j \phi_j = v(\{1, \dots, k\})$, the total reconstruction gain of the full model relative to the zero-factor reconstruction, so the normalized shares $\phi_j / \sum_l \phi_l$ form an additive decomposition of that gain; they sum to one by construction and are reported as percentages summing to 100%. Individual shares may be signed in principle but were all positive in the reported PDAC models. The reconstruction values $v(S)$ are uncentered, taken about zero as written above, and grand-mean centering enters only the separately reported centered R^2 of Cross-method reconstructability and reconstruction fidelity. These quantities describe different aspects of reconstruction and should not be interpreted as interchangeable measures. This is a genuine additive decomposition, in contrast to per-factor variance fractions, which for non-orthogonal factorizations such as NMF do not partition (cross-terms between rank-one contributions are non-zero). The Shapley shares are computed exactly by enumerating the 2^k subsets (trivial at the small ranks considered here).

The survival contribution of factor j was quantified as the Type III partial likelihood increment, computed on full-support training scores $W^\top X$ over all analyzed genes rather than on the truncated union support

used for external validation, as the reduction in Cox partial log-likelihood when factor j is omitted from the full k-factor model,

$$\Delta\ell_j = \ell_{\text{full}}(\hat{\beta}_{\text{full}}) - \ell_{-j}(\hat{\beta}_{-j}),$$

where $\ell_{\text{full}}(\hat{\beta}_{\text{full}})$ is the maximized unpenalized Cox partial log-likelihood of the full k-factor model and $\ell_{-j}(\hat{\beta}_{-j})$ is that of the reduced model refit on the remaining $k-1$ factors after omitting factor j, each evaluated at its own maximum partial likelihood estimate. Larger $\Delta\ell_j$ indicates a greater conditional contribution of factor j to Cox model fit given the remaining factors. Unlike the reconstruction Shapley shares, these Type III increments are conditional (each is evaluated against the full complement of the remaining $k-1$ factors) and are nonadditive: they do not partition the total model partial likelihood and are not comparable across factors as variance-style shares. We therefore report them as per-factor conditional increments rather than as an additive survival decomposition. These are post-fit unpenalized Cox comparisons, not differences obtained by deleting terms from the trained penalized DeSurv predictor. Because the gene programs were learned using these training outcomes, the increments describe the recovered representation and do not independently validate prognostic relevance.

These two quantities are plotted against each other in main text, Fig. 2c. In PDAC at $k = 3$, the reconstruction shares are 41.8%/18.6%/39.7% for NMF (N1/N2/N3) and 18.8%/25.2%/56.0% for DeSurv (D1/D2/D3), while the survival contribution $\Delta\ell$ concentrates in DeSurv D1 ($\Delta\ell = 66.4$) with all NMF factors small ($\Delta\ell \leq 1.2$).

10 Factor Correspondence Analysis

To assess which biological programs are preserved, altered, or suppressed under survival supervision, we quantified the pairwise correspondence between NMF and DeSurv factor loadings. For each pair of factors (j_{NMF}, j_{DeSurv}), we computed the Spearman rank correlation between the corresponding columns of the NMF gene program matrix W_{NMF} and the DeSurv gene program matrix W_{DeSurv} over all genes. We use the full loading profile rather than a top-gene subset, which would impose a threshold and overstate apparent preservation. Spearman correlation was used to capture monotone correspondence in gene loading ranks while remaining robust to scale differences between the two methods. Results are displayed as a heatmap with NMF factors as rows and DeSurv factors as columns (main text, Fig. 2d).

11 Analysis scripts and cached objects

Numerical results from the analyses are read at render time from cached analysis objects rather than transcribed manually. Table S5 indexes the scripts referred to in this document by the analysis they perform, so that the prose above and below can name analyses rather than files.

Part II

Supplementary Results

12 Cross-method reconstructability and reconstruction fidelity

The factor-correspondence heatmap of Section 10 (main-text Fig. 2d) summarizes pairwise similarity; to make a direction-aware statement of how much of each factor is *linearly reconstructable* from the other method's basis, we regressed each factor's gene-loading vector onto the full set of the other method's loading vectors and report the ordinary least-squares R^2 (Table S6). A high R^2 means the factor is well reconstructed as a linear combination of the other method's factors; a low R^2 means it is not linearly reconstructable from that basis. This is a statement about linear reconstructability of gene-loading vectors, not about which method discovered a program first.

DeSurv D1 is reconstructed at $R^2 = 0.09$ from the NMF basis: its gene-loading vector is not well reconstructed as a linear combination of the matched-rank NMF factors. The biologically dominant tumor (N1) and stromal (N3) programs, by contrast, are largely reconstructable from the DeSurv basis ($R^2 = 0.90$ and 0.88), so most of the unsupervised structure is retained; the principal reorganization is that the DeSurv basis resolves tumor signal into basal- and classical-aligned programs while reconstructing the exocrine factor (N2) less well ($R^2 = 0.38$).

At the whole-matrix level, this reorganization carries a real but bounded reconstruction cost (Table S7). On the conventional centered R^2, DeSurv reconstructs 0.75 of the analysis matrix against 0.81 for unsupervised NMF, a difference of 6.0 percentage points and a relative loss of about 7%; this centered figure is the one reported in the main text. On the uncentered scale the gap is only 1.5 percentage points, but that scale flatters any nonnegative low-rank fit to a rank matrix.

13 The D1 axis is not specific to DeSurv

If the D1 tumor–stroma axis were an artifact of DeSurv's particular optimizer, independent survival-supervised methods should not recover it. We therefore fit three other supervised dimension-reduction methods on the same training cohort ($n = 273$, 139 events), namely supervised principal components (Cox-screened genes followed by PCA), one-component Cox partial least squares against martingale residuals, and a sparse Cox model (lasso-penalized; 32 genes selected in this single fit), and compared their leading survival-aligned direction to the DeSurv D1 score. All three recovered D1 closely (per-patient $|r| = 0.81$ for supervised PCA, 0.83 for Cox-PLS, and 0.85 for sparse Cox; mean 0.83; Fig. S2), and each aligned with D1 rather than with the stromal D2 or basal-like D3 programs (absolute correlations with D2 and D3 at most 0.19).

By contrast, the leading *unsupervised* direction (the first principal component) was only weakly correlated with D1 ($|r| = 0.10$) and instead tracked the high-variance D2/D3 stroma–basal axis. This D1 axis is thus shared across the examined survival-supervised methods. Two additional findings are consistent with this conclusion: D1 is unreconstructable from the standard NMF basis (Table S6), and standard NMF at matched rank does not recover the restCAF and iCAF marker genes that appear among DeSurv's defining genes (Stromal resolution: which reference programs are recovered from bulk).

14 Robustness to tuned supervised comparators and compartment attribution

The preceding section establishes D1 as a survival-associated expression direction shared across supervised methods; here we ask what DeSurv adds beyond recovering that direction, using canonical, independently

tuned comparators in place of the simple single-component versions above: supervised principal components (`superpc`; Bair–Tibshirani, with cross-validated screening threshold and component number) and lasso-penalized Cox regression (`glmnet`, mixing parameter 1, a single $\lambda_{\min}$ model). Cox partial least squares was not included in this comparison because a validated multi-component implementation was not available. All methods used the same 1,970-gene universe selected from the training data by expression and variance, and each performed its own feature selection; for this comparison DeSurv's program scores used per-program masks, each column zeroed outside its own top-270 gene set, rather than the union support used elsewhere in the manuscript, and the Cox weights combining the three program scores were re-estimated without penalty on the training scores, so that all three methods enter as training-fitted linear scores on an equal footing. Validation outcomes were never used: all scoring used the frozen, rank-transformed transfer with no validation refitting.

External discrimination was similar: the pooled frozen-transfer C-index was 0.61 for the DeSurv linear predictor, 0.60 for supervised PCA, and 0.60 for lasso Cox. Consistent with the score-level recovery above, both tuned methods aligned with the D1 direction (correspondence between each supervised risk score and the DeSurv programs was quantified in the training cohort by absolute Pearson correlation; $|r| = 0.71$ for lasso Cox and 0.46 for supervised PCA) and with the DeSurv linear predictor ($|r| = 0.86$ for lasso Cox and 0.65 for supervised PCA), but were only weakly correlated with the activated-stroma/proCAF program D2 ($|r| = 0.06$ for lasso Cox and 0.14 for supervised PCA): a single supervised risk axis captures the dominant tumor-axis signal but not the separate stromal program.

The compartment attribution made this explicit. The genes emphasized by a supervised risk score need not coincide with D1's positively weighted defining genes, because the high-risk end of D1 is characterized in part by loss of classical-tumor and restCAF-like expression. Empirically they were basal-like tumor markers: across 12 cross-validation seeds the genes selected by both methods were enriched (hypergeometric $P < 0.01$) for the basal-like tumor program (in 12/12 seeds for lasso Cox and 12/12 for supervised PCA) and in no seed for the activated-stroma/proCAF, restraining-stroma, immune, or exocrine compartments. The supervised risk scores did nonetheless retain some stromal signal: each correlated modestly with an external activated-stroma/proCAF signature ($|r| = 0.20$ for lasso Cox and 0.23 for supervised PCA), but well below DeSurv's D2 program ($|r| = 0.55$), which tracks that signature directly.

The distinction is therefore representational: DeSurv preserves stromal context as an explicit program even when that program does not contribute independently to the fitted linear predictor. The independently tuned methods used different feature-selection procedures and retained different numbers of genes, and neither mechanism matches DeSurv's. The penalized Cox comparator uses a pure lasso penalty (mixing parameter 1, distinct from DeSurv's supervision parameter α), whereas supervised principal components applies no penalty at all, instead screening genes by a cross-validated feature-score threshold and extracting a cross-validated number of components; DeSurv's internal Cox step uses an elastic net whose Bayesian-optimization-selected mixing parameter ($\xi = 0.056$) sits close to ridge. Both comparators retain far fewer genes than DeSurv's 810-gene support: across 12 resampling seeds, lasso Cox selected a median of 14 genes (range 9–33) and supervised principal components 39 (range 5–222), and both were tuned by their standard method-specific cross-validation rather than a budget matched to DeSurv. Absence of reference-program enrichment in their selected gene lists does not establish absence of stromal information in their risk scores.

15 Proper-scoring sensitivity of the external validation

The main-text external validation reports discrimination with Harrell's C-index under a Cox proportional-hazards model. Harrell's C ranks patients by a single time-invariant score, so when hazards are non-proportional, as the diagnostic below shows for this predictor, its value depends on the censoring distribution and need not reflect discrimination at any particular horizon[28,29], and it is not a proper scoring rule for predicted risks[30]. We therefore additionally evaluated the DeSurv linear predictor, its direction frozen from training, with a proper score, the integrated Brier score (IBS), in each validation dataset, recalibrating only its scalar slope and baseline hazard within each dataset and integrating over that dataset's follow-up to the 80th percentile of its event times.

Across the five validation platform datasets (PACA-AU array restricted to its array-only patients) the DeSurv predictor achieved an event-weighted integrated Brier score of 0.159 versus 0.168 for a Kaplan–Meier reference, an index of prediction accuracy (IPA $= 1 - \mathrm{IBS}/\mathrm{IBS}_{\mathrm{null}}$) of 5.4% (per-dataset range 2.9–9.4%). Because each dataset's IBS is integrated over its own event-time horizon, this pooled value is a descriptive event-weighted summary rather than a single common-horizon proper score, and the per-dataset IPA range is the more directly interpretable quantity.

Two caveats limit the interpretation of this analysis. First, the recalibration step estimates both the scalar slope and the baseline hazard from each validation dataset's own outcomes, so unlike the C-index and hazard-ratio analyses, which use a fully frozen predictor, this sensitivity evaluates the transported score direction under local recalibration rather than a frozen absolute-risk model. Second, the scores are computed on the same data used to fit that recalibration, with no cross-validation or resampling, so the reported IBS and IPA are apparent in-sample values that are subject to optimism and do not estimate out-of-sample accuracy. Both caveats weaken the evidentiary value of the apparent improvement, which is already small. This modest proper-score improvement over the null is consistent with the moderate discrimination (C = 0.61) and with DeSurv's contribution being a separately resolved tumor- and stroma-associated decomposition rather than higher predictive accuracy.

16 Proportional-hazards diagnostic

Because the pooled hazard ratio assumes proportional hazards, we tested the scaled Schoenfeld residuals of the validation model, a Cox regression of overall survival on the standardized DeSurv linear predictor with a separate baseline hazard for each dataset (stratification), using the Grambsch–Therneau test. The proportional-hazards assumption was not supported for the standardized DeSurv linear predictor ($\chi^2 = 14.7$, df $= 1$, $P = 1.3 \times 10^{-4}$), with the association attenuating over follow-up: the estimated time-varying coefficient declines with follow-up (slope -0.92 on Kaplan–Meier-transformed time) from a positive value. The reported pooled hazard ratio is therefore a time-averaged summary of that association, not a constant multiplicative effect. The horizon-integrated proper-scoring (IBS/IPA) sensitivity above complements it by integrating prediction accuracy over each cohort's event-time horizon rather than through a single hazard-ratio summary.

17 Between-cohort heterogeneity and C-index confidence intervals

The five validation platform datasets span four independent patient cohorts, because the PACA-AU array and RNA-seq series profile a partially overlapping set of the same ICGC patients. All analyses in this section are patient-level and de-duplicated by the rule used throughout the manuscript: a patient profiled on both PACA-AU platforms is represented once, by the RNA-seq record, leaving 69 PACA-AU patients (17 array-only and 52 RNA-seq) and 570 unique patients with 388 deaths in total.

The DeSurv linear predictor was standardized *within* cohort (centered and scaled to unit standard deviation separately in each cohort), the same within-group convention as the per-factor forest estimates in main-text Fig. 3a, there applied to the five platform datasets, and not the pooled-standard-deviation convention used for the headline pooled hazard ratio (Survival Analysis). Within-cohort standardization removes the scale degree of freedom (every cohort's score then has mean zero and unit standard deviation by construction), so a cohort-by-score interaction tests only whether the slope per within-cohort standard deviation differs between cohorts, and not whether the score is more dispersed in one cohort than another. Both models are Cox regressions of overall survival with a separate baseline hazard for each cohort (stratification): the common-slope model has the standardized predictor as its single covariate, and the cohort-specific-slope model adds the predictor-by-cohort interaction. Comparing them by likelihood-ratio test gave $\chi^2 = 1.47$ on 3 degrees of freedom ($P = 0.69$); standardizing within platform rather than within cohort, which removes the small residual location difference between the two PACA-AU platforms, gives $\chi^2 = 1.59$ on 3 degrees of freedom ($P = 0.66$).

Cochran's Q over the four cohort-specific log hazard ratios and their standard errors was 1.47 on 3 degrees of freedom ($P = 0.69$), giving $I^2 = 0\%$ under $I^2 = \max\{0, (Q - \mathrm{df})/Q\}$. Q and I^2 are descriptive summaries of dispersion among the four estimates and are reported as complements to the interaction test, not as a substitute for it. The likelihood-ratio test is the inferential statement, and with four cohorts both are low-powered against modest heterogeneity.

Confidence intervals for the C-index were obtained by a stratified nonparametric bootstrap on the same de-duplicated table, resampling patients with replacement *within* cohort so that no replicate mixes patients across cohorts or reinstates a de-duplicated PACA-AU record (B = 1,000 replicates, seed 1, percentile intervals). Concordance was evaluated as Harrell's C with higher scores treated as higher risk, the convention of the external-validation C-indices, and the heterogeneity script (Table S5) verifies that this reproduces the per-dataset C-indices of Supplementary Table S10 exactly. The combined value is the event-weighted mean of the cohort C-indices, the same convention as the proper-scoring and tuned-comparator analyses, and therefore never forms a concordance pair across cohorts. The combined C-index was 0.613 (95% CI 0.579–0.646); the per-cohort hazard ratios and C-indices are given in Table S11.

17.1 Leave-one-dataset-out sensitivity of the pooled hazard ratio

18 Variance partition of the D1 score

The main text reports the fraction of the D1 score's variance explained by the binary PurIST and DeCAF classifications. D1 scores are first standardized within platform dataset (five levels, applied after PACA-AU de-duplication, so the array and RNA-seq platforms remain separate units), and an ordinary least-squares model is then fitted with platform dataset (five levels) as a fixed effect together with PurIST, DeCAF and their interaction. The quantity reported is the ordinary R^2 of that model, not the adjusted R^2; on $n = 570$ with eight estimated parameters the adjusted value is about one percentage point lower. Because the scores are standardized within platform dataset before fitting, the dataset main effect absorbs essentially no variance, so the within-dataset normalization of the reported statistic is a formality rather than a correction. The interaction term contributes negligibly, so the additive and interaction models give the same figure to the reported precision. Prognostic increments over the classifiers are assessed separately, by a partial likelihood-ratio test and the change in concordance from adding D1 to a Cox model stratified by platform dataset (five strata) containing both classifiers.

D1 could also be a smoothed coding of the four joint PurIST-by-DeCAF subtype cells, so that its classical and restCAF-like arms would simply re-express one cell of a two-by-two. The main text reports the additive adjustment (D1 added to a Cox model stratified by platform dataset containing PurIST and DeCAF as main effects). As a sensitivity analysis we repeated that test with the PurIST-by-DeCAF interaction also in the base model; the statistics below are from this interaction model and differ slightly from the main-text additive ones for that reason. The interaction term itself is not prognostic (HR 1.42, $P = 0.28$), so there was no evidence that the joint classification improved prognostic fit over the additive one. Adding D1 does improve it: it gives a partial likelihood-ratio statistic of 10.44 on 1 degree of freedom ($P = 0.0012$), an adjusted hazard ratio per standard deviation of 0.82 (0.73–0.92), and a concordance increase from 0.592 to 0.621. Re-coding the two classifiers as a single saturated four-level categorical variable is the same model reparameterized and returns the identical statistic ($\chi^2 = 10.44$). D1's prognostic information is therefore not captured by the saturated four-level joint classification.

19 Adjustment for clinical covariates and tumor purity

The association of the frozen DeSurv linear predictor with survival was estimated in cohort-specific Cox models before and after adjustment for the clinical covariates that cohort annotates and, where available, for tumor purity (Table S9). Purity annotation is present in only a subset of cohorts and is derived by different estimators across them (TCGA-PAAD purity was estimated by ABSOLUTE; the PACA-AU and Moffitt

purity fields were used as distributed with the cohort annotations, which do not identify the estimator), so it is used here solely as an adjustment covariate; wherever purity could be measured on a full cohort the adjusted hazard ratio was essentially unchanged from the unadjusted one, providing no evidence that the association was explained by measured tumor purity. Consistently, the D1 score itself correlated only weakly with purity in the two full-cohort datasets (Spearman $\rho = 0.06$ in PACA-AU array and -0.05 in PACA-AU RNA-seq), the values reported in the main text.

20 Empirical optimization stability across random initializations

The idealized alternating scheme is not accompanied by a formal convergence guarantee for the fitted models. With $\lambda_H = 0$, as in the reported analyses, strict convexity of the H-subproblem and uniqueness of its block minimizer are not guaranteed a priori, so convergence results that require unique block minimizers do not apply directly; in addition, the implemented algorithm combines column normalization, rescaling of the weighted Cox gradient (the reconstruction terms are not rescaled) by the norm-based factor ζ, subject to its stabilizer and cap, and a Coxnet approximation of the β update. We therefore assessed optimization behavior empirically across $R = 100$ independent random initializations of the selected PDAC model, with a stopping tolerance of 10^{-7} applied to the cosine change of W and of H between successive outer iterations (a rejected W step counts as zero change, so the rule can be met by H converging with W fixed; Optimization Algorithm), and a maximum of 4,000 iterations.

100 of the 100 runs terminated on the stopping criterion rather than the iteration cap, none failed numerically, and in every run the recorded outer-loop objective decreased monotonically; the median number of iterations was 38 (range 17–233). The recorded series is $\mathcal{L}$ as defined in Model details, evaluated once per outer iteration after the β update. It ends with the last outer iteration and does not include the post-loop β refit, whose objective is stored separately (Update for β), so monotonicity here is a statement about the outer loop. The size of the decrease from a random start varies widely between initializations, but that quantity primarily reflects the starting objective value rather than agreement among the final solutions. Whether independent runs of the reported procedure recover the same gene programs is addressed in Stability of the recovered gene programs below.

21 Stability of the recovered gene programs

The previous section documents that every run stops on its criterion with a monotonically decreasing objective. It does not establish that the programs it reaches are the same ones each time, and that is the property the manuscript relies on when it applies a frozen decomposition across cohorts. We therefore measured stability at the level of the recovered programs rather than the objective, using the same $R = 100$ cached initializations. No model was refitted for the restart analysis.

Individual restarts are highly dispersed. Comparing pairs of independent random restarts, and matching their columns one to one by absolute Spearman correlation of gene loadings, the median overlap between the top 270 genes of matched programs was 0.13 by Jaccard index, against a chance baseline of 0.074 for two independent draws of that size from 1,970 genes. Matched gene-loading correlations were moderate (0.59–0.60 across the three programs). Over the same runs, training concordance was nearly constant (median 0.704, standard deviation 0.010). Near-identical predictive performance from bases that share little gene membership is consistent with a non-identifiable objective, and it is precisely why the reported model is not a single run.

The reported procedure is stable to restart-pool composition within the observed training data. The reported basis is obtained by aggregating restarts into a consensus initialization and then fitting from it. To test whether that procedure returns the same programs, we partitioned the cached restarts into two disjoint halves, built a consensus initialization from each half independently, fitted from each at the production settings, and compared the results (8 random splits). Across the eight random splits, the paired consensus fits

agreed closely with each other (gene-loading correlation 0.93–0.95, patient-score correlation 0.97–0.99), and each agreed still more closely with the reported basis (gene-loading correlation 0.97–0.99, top-270 Jaccard 0.73–0.88). Each half contained 50 of the same 100 cached starts and therefore shared restarts with the pool used to construct the reported basis. These comparisons assess sensitivity to restart-pool composition within the observed training data; they are not independent repetitions of the full procedure. Using fewer restarts does not establish that the agreement estimates are conservative.

The fitted basis also moved materially from the consensus initialization W_0: with programs corresponding by position (confirmed against the full correlation matrix), gene-loading correlations between W_0 and the final W were 0.57–0.81 and top-270 Jaccard overlaps were 0.44–0.82. The consensus therefore supplies a reproducible initialization rather than fixing the final basis.

The optimizer alone is not identifiable. The consensus step is what makes the reported decomposition stable to restart-pool composition rather than one draw from a wide distribution, and reproducibility claims in this manuscript refer to that end-to-end procedure.

Stability under data perturbation is a separate and weaker property. The analyses above vary the initialization while holding the data fixed. Refitting on 25 repeated 80% subsamples of the training cohort, with each DeSurv refit a single run from one random initialization at the production hyperparameters rather than the 100-restart consensus procedure, and comparing the top-100 gene sets of the D1-matched factor across subsamples instead gives a mean pairwise Jaccard of 0.34 for DeSurv, against 0.49 for Cox partial least squares, 0.38 for supervised principal components and 0.22 for sparse Cox. DeSurv is mid-range. These figures come from the frozen-axis comparator analysis, which works on the full shared support of genes common to the trained basis and the analyzed expression matrix rather than on the truncated union, and whose frozen-transfer linear predictor refits unpenalized Cox coefficients on the three training scores (Survival Analysis). The 100-gene list is shorter than the 270-gene projection support so that the comparison is on a common footing with the single-vector comparators, whose gene rankings are not factor-specific. This comparison favors dense methods by construction, because a score that spreads weight across many genes has a more stable top-100 list than one built from a selective basis, as the sparse Cox value illustrates; it also disfavors DeSurv relative to the consensus procedure used for the reported model, because each refit here is a single run. Gene-level resampling stability addresses a complementary property distinct from reproducibility of the frozen end-to-end procedure.

22 Simulation design and data generation

Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from Gamma(shape $= 3$, rate $= 0.8$) and cross-loadings onto other factors from Gamma$(1, 20)$; background gene loadings from Gamma$(2, 1)$ across all factors; and noise gene loadings from Gamma$(1, 20)$. Sample-level factor activities H were drawn from Gamma$(2, 2)$. Each dataset comprised $p = 3{,}000$ genes, $n = 200$ samples, and $k = 3$ factors, with 150 marker genes per factor and 500 shared background genes. The three factors are exchangeable in this design: the prognostic factor is not generated with lower variance than the other two. The variance-dominant signal excluded from the survival-generating gene set is the shared background block, which loads on every factor, so the factor-specific marker genes that carry survival contribute less of the total expression variation than that block. Because the background genes load on the same latent activities as the markers, their expression remains correlated with the linear predictor; "background" here denotes exclusion from the survival-generating gene set, not statistical independence of outcome. The expression matrix actually analyzed was not the noise-free product WH: a per-gene baseline drawn as $\mathcal{N}(3, 1)$ was added to every row, independent Gaussian noise with standard deviation 0.01 was added to every entry, and negative values were clipped to zero.

Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic and mixed scenarios and $\beta = 0$ in the null scenario, which otherwise used the same baseline hazard parameter and censoring rate as the prognostic

scenario. The mixed scenario differs from the prognostic scenario only in its survival gene set, 300 genes per column of the survival-generating weight matrix W^*, comprising the 150 markers of that factor and 150 randomly sampled background genes, which creates partial overlap between the prognostic signal and the variance-dominant background.

Survival times were generated from an exponential distribution, from the noise-free expression only. Let $X_0 = WH$ denote that noise-free matrix and W^* the survival-generating weight matrix, a masked copy of the column-normalized W that retains, in each column, only the loadings of that factor's survival-generating genes and is zero elsewhere: the 150 markers of that factor in the prognostic and null scenarios, and 150 markers plus 150 background genes in the mixed scenario. The raw generating scores were $Z_{\text{raw}} = X_0^\top W^* \in \mathbb{R}^{n \times k}$, written here as the transpose of the main-text model convention $Z = W^\top X$ so that rows index samples, and Z_{true} denotes their column-wise standardization across all $n = 200$ samples, applied before the 70/30 train/test split; the linear predictor was $\eta = Z_{\text{true}}\beta$, so with $\beta = (2, 0, 0)^\top$ only factor 1 contributes; event times were drawn as $T \sim \text{Exponential}(\lambda_0 e^\eta)$ with $\lambda_0 = 0.05$, and censoring times independently as Exponential(0.02). Each scenario comprised 100 replicates, each with a held-out test split.

Recovery of the prognostic gene set was summarized by matched-factor precision, defined in the main-text Methods, and by global marker recall, defined below in Simulation results under null and mixed scenarios. The ground-truth target against which both are scored is scenario-specific. In the prognostic scenario the target comprised the 150 factor-specific marker genes driving survival ($\beta = (2, 0, 0)$); in the mixed scenario, marker-only and full survival-gene definitions of the target were evaluated separately.

23 Simulation results under null and mixed scenarios

To characterize the operating characteristics of DeSurv across different signal regimes, we evaluated performance under three simulation scenarios. In the prognostic scenario (main text Fig. 5), the factor-specific survival signal contributed relatively little transcriptomic variation compared with the shared background signal excluded from the survival-generating gene set, and the specified survival-generating gene set comprised 150 factor-specific marker genes ($\beta_1 = 2$, $\beta_2 = \beta_3 = 0$). In the null scenario, survival times were generated independently of gene expression, with generating coefficients $\beta_1 = \beta_2 = \beta_3 = 0$, so no prognostic structure existed. In the mixed scenario, 300 genes per column of the survival-generating weight matrix W^* entered the survival gene set, each column carrying 150 factor-specific markers and 150 background genes with loadings shared across all factors, with survival weight placed on factor 1 only ($\beta_1 = 2$, $\beta_2 = \beta_3 = 0$). All scenarios used $p = 3{,}000$ genes, $n = 200$ samples, $k = 3$ true factors, the same baseline hazard parameter (0.05) and the same independent exponential censoring rate (0.02), and identical gamma distribution parameters for the gene loading matrix W (Simulation design and data generation). In every scenario the generating scores are formed from the noise-free expression matrix $X_0 = WH$ rather than from the analyzed matrix. The raw generating scores are $X_0^\top W^*$, and their column-wise standardization across all $n = 200$ samples, applied before the train/test split, gives the standardized scores from which the linear predictor is formed, so the generating β is per standard deviation of the generating score and is not comparable to a coefficient on a raw projection. Performance was evaluated using the concordance index (C-index) on the held-out 30% test split of each replicate, precision of recovered prognostic gene sets, and the distribution of selected ranks across simulation replicates. In the prognostic scenario the shared background genes are excluded from the survival-generating gene set; their exclusion does not imply statistical independence from survival, because background and marker expression share latent factor activities.

In the prognostic scenario, median matched-factor precision was 0.50 for DeSurv versus 0.07 for unsupervised NMF, and global marker recall, the fraction of the survival-driving marker genes recovered anywhere in the union of all learned factors' top-n_{top} gene sets, was 0.26 versus 0.15. Matched-factor precision is optimistic by construction, because the matched factor is selected using the same precision that is then reported; global marker recall, which searches the union of all learned factors, is the matching-free complement. Like precision, it depends on each method's selected n_{top}. It complements the matched-factor precision reported in main-text Fig. 5.

Under the null scenario (Fig. S3a–b), DeSurv showed no advantage over unsupervised NMF ($\alpha = 0$): both methods yielded C-index values near 0.5, and selected ranks concentrated at low values (NMF favoring $k = 2$), consistent with the absence of survival signal. Because the survival objective is uninformative under the null, the cross-validated concordance is essentially flat in α and the Bayesian-optimized supervision strength is unidentified, dispersing across its admissible range rather than concentrating near zero (median $\alpha = 0.59$, IQR 0.19–0.82; range 0–0.94 across 100 replicates). The null-scenario specificity thus reflects the flat concordance surface itself, not the optimizer driving α to zero.

Under the mixed scenario (Fig. S3c–f), DeSurv's advantage was concentrated in marker-gene recovery. Precision against the factor-specific marker genes alone (150 genes; Fig. S3d left) substantially favored DeSurv, whereas precision against the full 300-gene survival set including background genes (Fig. S3d right) was more modestly improved, indicating similar background-gene recovery between methods. The learned factor aligned with the true marker program also received a larger Cox coefficient ($|\beta|$) under DeSurv (Fig. S3e), and C-index was modestly higher (Fig. S3c). Across the three scenarios the pattern was graded: DeSurv's advantage was greatest when variance and prognosis diverged (prognostic), attenuated in full-target precision and in discrimination when they partially overlapped (mixed, as background-gene dilution reduced the discrimination gain) while marker-only precision still favored DeSurv, and absent when no survival signal existed (null). These simulations indicate that DeSurv is most advantageous when prognostic structure is not aligned with the dominant sources of transcriptomic variation.

24 Cross-validated C-index across factorization rank

To evaluate how the number of latent factors k affects prognostic performance, as a sensitivity analysis we performed an exhaustive grid search over $k \in \{2, 3, \ldots, 12\}$ and supervision strength $\alpha \in \{0, 0.05, \ldots, 0.95\}$ using 5-fold cross-validation on the TCGA+CPTAC training cohort. For each k, the best-performing α was selected by maximizing the mean CV C-index. The unsupervised limit ($\alpha = 0$) was evaluated at the same k values for comparison. Results are shown for two gene-selection settings: $n_{\text{top}} = 270$ top genes per factor and $n_{\text{top}} = \text{ALL}$ (all genes retained), both at $\lambda = 0.349$ and $\xi = 0.056$. External validation C-index is shown descriptively; validation outcomes were not used to select k, α, λ, ξ, or n_{top}.

Fig. S4 shows the resulting C-index curves for both training (CV) and external validation (pooled across held-out cohorts), with shaded ribbons indicating ± 1 standard error. Rank was selected on the training cross-validated concordance. For DeSurv that curve is close to flat across the whole range examined, spanning about two standard errors from $k = 2$ to $k = 12$, with its nominal maximum at $k = 9$ rather than at the selected rank. This fixed-λ, ξ, n_{top} grid is a different exercise from the full Bayesian-optimization search over all tuned hyperparameters, whose highest observed mean C-index occurred at $k = 7$. Applying the same one-standard-error rule to panel (a) of this sensitivity grid, at $n_{\text{top}} = 270$, also selects $k = 3$, from a grid best of 0.671 (standard error 0.009) at $k = 9$ and the resulting threshold 0.662. That threshold is the grid's own, distinct from the Bayesian-optimization threshold of 0.644, and what it corroborates is the selected rank rather than the exact final configuration, whose nearest neighbor in this grid clears only the Bayesian-optimization threshold (Bayesian Optimization). The selection depends on the gene-selection setting: under panel (b), with all genes retained, the same rule selects $k = 2$. On the training cross-validated curve DeSurv had a higher point estimate than its unsupervised limit at every k examined. The external-validation curve, shown only descriptively, does not show that ordering uniformly: the unsupervised limit is higher at $k = 6$ and at $k = 10$–12, and the supervised curve dips at $k = 6$, where the single grid fit at its selected α is the same fit whose factors correspond to almost no reference program (Reference-program coverage, edges and multiplicity). These rank-wise points are single fits, not consensus fits. At the selected rank the validation curve is consistent with the training-based choice, with DeSurv at $k = 3$ attaining concordance comparable to the rank-optimized unsupervised control ($k = 7$; Table S10).

25 Standard NMF rank selection heuristics

Standard unsupervised rank selection criteria (reconstruction residuals, cophenetic correlation coefficient, and mean silhouette width) yielded inconsistent guidance for selecting the factorization rank k in the PDAC training cohort (TCGA + CPTAC). Reconstruction residuals decreased smoothly without a visually distinct elbow (Fig. S5a); the Kneedle elbow rule (maximum deviation of the normalized curve from its chord) applied to this curve nonetheless returns $k = 5$, which is the rank used for the elbow-selected standard NMF comparator in Table S10. Cophenetic correlation fluctuated without a distinct transition point (Fig. S5b). Mean silhouette width favored very low ranks that conflicted with other criteria (Fig. S5c). These inconsistencies illustrate the ambiguity of rank selection under unsupervised criteria in this dataset; the main text selects rank on cross-validated prognostic performance.

26 Factor structure of the rank-optimized unsupervised control ($k = 7$)

Throughout the main text, both DeSurv and standard NMF were fit at the same rank ($k = 3$), enabling a controlled comparison of the effect of survival supervision on factor structure. When the unsupervised limit of the same pipeline selects its own rank, the one-standard-error rule applied to the $\alpha = 0$ Bayesian-optimization results selects $k = 7$ (the rank-optimized unsupervised control), whereas the same rule applied to the supervised DeSurv results selects $k = 3$. We therefore examine the factor structure of this independently selected $k = 7$ solution to characterize how the additional factors relate to established programs when supervision is absent (Fig. S6).

At BO-selected $k = 7$ under $\alpha = 0$ (Fig. S6), the rank-optimized unsupervised control corresponds to a broader set of PDAC gene signatures than the matched-rank $k = 3$ solution, with some programs represented by more than one factor; Supplementary Fig. S7 quantifies the accompanying changes in coverage and multiplicity.

26.1 Reference-program coverage, edges and multiplicity

Fig. S6 (and main-text Fig. 2a,b) display factor-to-reference-program correspondence as a heatmap. We additionally summarize this same correspondence across factorization rank k using three quantities computed from the same correlation matrix (top 50 genes per factor by factor-specificity score, Spearman correlation of factor loadings against binary reference-program membership, $\rho > 0.2$ representation threshold; Supplementary Information, Section 8), over the fixed 45-program reference universe. Coverage is the number of reference programs with at least one factor at $\rho > 0.2$. Edges is the total number of (factor, reference-program) pairs at $\rho > 0.2$. Reference-program multiplicity is edges divided by coverage, defined only when coverage > 0, and reports edges per represented program. A low multiplicity is not by itself evidence that fewer programs are represented, so multiplicity is always reported alongside coverage rather than in place of it.

At each rank $k = 2, \ldots, 12$, these quantities are computed for two single fits at fixed λ, ξ and $n_{\text{top}} = 270$ from the $k \times \alpha$ cross-validation grid: the unsupervised fit ($\alpha = 0$) and the fit at the α maximizing mean cross-validated C-index, the same α-selection convention used for Supplementary Fig. S4. These rank-wise points are single fits at fixed hyperparameters, not the 100-restart consensus fits used elsewhere in the manuscript. Three final-model anchors are overlaid: the consensus fits for DeSurv ($k = 3$) and the rank-optimized unsupervised control ($k = 7$, $\alpha = 0$), and the best-of-30-run `NMF::nmf` fit at matched rank ($k = 3$).

At these anchors, DeSurv reaches coverage 16, edges 16, multiplicity 1.00; matched-rank standard NMF reaches coverage 15, edges 15, multiplicity 1.00; and the rank-optimized unsupervised control at $k = 7$ reaches coverage 32, edges 36, multiplicity 1.125.

The supervised curve collapses at two ranks: at $k = 6$ ($\alpha = 0.1$) and $k = 8$ ($\alpha = 0.2$) the single grid fit reaches coverage 2 and 2 (edges 2 and 3), because the largest correlation between any of its factors and any reference

program is only 0.21 and 0.29, barely above the 0.2 representation threshold, whereas the unsupervised fits at the same ranks and the supervised fits at the neighboring ranks $k = 5, 7, 9$ all reach at least 0.57. These are single fits from one random start at fixed hyperparameters, and the $k = 6$ fit is also the one whose external concordance dips in Supplementary Fig. S4; these low-coverage endpoints are consistent with the single-start variability documented in Stability of the recovered gene programs; consensus fits at these ranks were not evaluated by this curve. Across the remaining ranks, coverage increases with rank considerably more than multiplicity does, indicating that the additional factors at higher rank extend correspondence to more reference programs rather than densely duplicating correspondence to the same ones. Through $k = 7$ the dominant change is increased coverage with modest repeated representation. This is not a claim about predictive performance, which is addressed separately (Table 1; Supplementary Table S10).

26.2 Stromal resolution: which reference programs are recovered from bulk

Bulk deconvolution has conventionally separated activated from normal stroma, whereas the finer cancer-associated fibroblast states, and the proCAF/restCAF classification derived from them, were established in single-cell data. We therefore asked which stromal reference programs each $k = 3$ factorization actually resolves. Two views are reported because they do not agree, and the defensible claim depends on which question is asked.

In the top-gene view shown in main-text Fig. 2a,b, which asks whether signature genes are represented among the 50 genes that define each factor, the activated-stroma program is recovered comparably by all methods (7, 6 and 10 of 206 genes for DeSurv, standard NMF and the matched-budget control). The restraining-stroma programs behave differently: 2 of 25 restCAF genes and 2 of 35 iCAF genes appear among DeSurv's defining genes, against 0 and 0 for standard NMF. The proCAF programs, by contrast, are recovered comparably by all methods (4 versus 3 of 25 genes).

In the full-gene view, which correlates loadings across all 1,970 analyzed genes with signature membership, no method separates restCAF from proCAF: every correspondence lies between 0.04 and 0.10, whereas the activated-stroma axis is recovered by all methods at 0.19. This view is far more dilute for a 15–25 gene signature among 1,970 genes and correspondingly low powered, but it bounds what can be claimed. We therefore report the restraining-stroma difference as marker-gene presence among the defining genes of each factorization, and do not claim to have resolved the single-cell CAF states from bulk expression.

Note also that the SCISSORS iCAF and myCAF reference signatures are displayed as restCAF and proCAF in main-text Fig. 2, following the naming used by the DeCAF classification; the underlying gene sets are the single-cell-derived ones.

26.3 Matched-budget unsupervised control at the same rank

The matched-rank comparison in the main text uses Lee multiplicative NMF (`NMF::nmf` with `nrun = 30`, which returns the single run of the thirty with the lowest reconstruction residual), which receives neither DeSurv's consensus initialization nor its hyperparameter tuning. That asymmetry leaves open whether the difference in factor structure reflects tuning effort rather than supervision. We therefore fit an unsupervised factorization control matched to DeSurv on rank, optimizer, search budget, gene filter, cross-validation folds, consensus procedure and stopping settings. With α fixed at 0 and the rank fixed at DeSurv's selected $k = 3$, the remaining hyperparameters were re-optimized within this otherwise identical pipeline, and the resulting model was fit by the same 100-restart consensus procedure used for DeSurv. This comparison isolates removal of survival supervision at the pipeline level rather than holding all non-supervision hyperparameters numerically fixed; the sensitivity analysis below holds DeSurv's λ, ξ and n_{top} fixed and varies only α. Because both $\alpha = 0$ controls are fit by the DeSurv optimizer, they share its stopping settings (tolerance 10^{-7} on the cosine change of W and of H, at most 4,000 outer iterations), whereas `NMF::nmf` stops on its own connectivity-based stationarity criterion (40 consecutive checks without a change in the connectivity matrix) under a 2,000-iteration cap.

The control selected $\lambda = 0.0025$, $\xi = 0.0566$ and $n_{\text{top}} = 299$; note that its gene support was selected near the upper bound of the search range (300), whereas the supervised model settled at $n_{\text{top}} = 270$ well inside it. At $\alpha = 0$, λ and ξ govern the post hoc Cox fit rather than entering the factorization objective. These are tuning-selected values and not the values used for external scoring. Both $\alpha = 0$ controls were scored on the union of their own factors' top-gene sets at the supervised model's common $n_{\text{top}} = 270$ rather than at their own tuned 299 (this control) and 187 (the rank-optimized $k = 7$ control), so the primary factorization comparison used a common number of top genes per program when constructing each model's scoring union; total union size can still differ with rank and overlap among programs. Scoring this control at its own tuned value instead changes no per-dataset C-index by more than 0.007 (Supplementary Table S10).

The control does not recover D1. Its best-matching factor reaches Spearman $\rho = 0.24$ against D1, slightly lower than the $\rho = 0.36$ attained by `NMF::nmf` at the same rank, while still recovering D2 ($\rho = 0.86$) and D3 ($\rho = 0.75$). Its external discrimination is likewise no better than `NMF::nmf` at $k = 3$ (Table S10). A sensitivity analysis holding the supervised model's own λ, ξ and n_{top} fixed and varying only α gave the same conclusion. Matching the tuning budget therefore does not close the representational gap, supporting survival supervision rather than tuning effort alone as the source of the reorganization reported in the main text.

Standard NMF at elbow-selected $k = 5$ and the rank-optimized unsupervised control at $k = 7$ both improve over standard NMF at $k = 3$ in all five validation datasets (Table S10); additional unsupervised factors capture prognostically relevant variation missed at the matched rank. At $k = 7$, the rank-optimized control exceeds DeSurv's C-index in all five validation datasets, so unsupervised factorization given more factors discriminates at least as well as the supervised three-program model. The comparison also differs in representation size: the $k = 7$ control represents a broader set of established programs across seven factors (Fig. S6; Supplementary Fig. S7) rather than concentrating the prognostic signal in three programs. Both the $k = 7$ control and the DeSurv linear predictor retain significant prognostic value after adjustment for established classifiers (main-text Table 1); DeSurv does so with fewer factors. Both models selected their rank by the same one-standard-error rule applied to their own Bayesian-optimization results; seven is the smallest rank at which the $\alpha = 0$ search came within one standard error of its best configuration (which had rank 11), not a general minimum rank for unsupervised recovery of comparable prognostic information.

27 Supplementary Data: factor-specific gene lists

As Supplementary Data, Tables S12–S14 list the top 270 genes for each of the three DeSurv factors, ranked by the factor specificity score s_{ij} defined in Supplementary Information, Section 8. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ($n_{\text{top}} = 270$); these are the genes used to compute sample-level projected scores for survival prediction.

Gene symbols in these tables are retained as distributed with the source expression matrices compiled for the training and validation cohorts (Section 6) and were not updated to current HGNC nomenclature; several therefore appear under previous symbols (for example SEPT11, WISP1, CYR61 and H1F0). Because the analysis set was selected for high mean expression as well as high variance, abundantly expressed ribosomal-protein and housekeeping genes are eligible for the factor supports, and some appear among the D2 and D3 defining genes (for example RPS26 and RPS28 in D2; RPS16, RPS21, RPL28 and GAPDH in D3); they enter the supports through the factor-specificity ranking and are not excluded post hoc. In the fitted model, each factor had at least 270 genes with positive factor specificity (284, 401, 1,285 for D1, D2 and D3). Because this specificity can be positive for at most one factor, the three top-270 sets were disjoint, yielding an 810-gene projection support. Factor D1 is enriched for classical tumor and restCAF-like stromal genes, Factor D2 for proCAF-associated stromal genes, and Factor D3 for basal-like tumor genes.

28 Translational transportability into independent treated PDAC cohorts

The programs were learned in treatment-naive, non-metastatic PDAC, the setting in which the gene-program matrix W and Cox coefficients β were trained. As a translational test we projected the fixed programs, without retraining, into independently treated PDAC and asked whether they retained clinically relevant associations. We used two resources. The first is the O'Kane/COMPASS cohort of treated metastatic PDAC, laser-capture microdissected and therefore epithelial-enriched, which supports a transportability test of the tumor-associated D_1 direction under a model stratified by treatment arm (Fig. 4a). The second is the borderline-resectable or locally advanced Linehan cohort, in which paired pre- and post-treatment biopsies allow a within-patient test of whether the projected D_2 score changes with therapy (Fig. 4b).

Projection used the frozen union support without any retraining. Expression was rank transformed within sample over the genes each cohort shares with the trained gene-program namespace; support genes absent from a cohort were omitted from that cohort's projection rather than imputed (805 of the support genes were measured in the COMPASS data), and the resulting program scores were standardized within each treated cohort before analysis. Because the omission rule drops rather than imputes missing support genes, a projected score is comparable within a treated cohort but not on the same absolute scale as the training-cohort score.

Each analysis is confined to a single cohort; no estimate is pooled across cohorts. Within O'Kane/COMPASS, arm-specific estimates are reported alongside a common D_1 slope from a Cox model stratified by treatment arm, which allows arm-specific baseline hazards. The elastic-net penalty shrank the trained survival coefficient for D_2 to exactly zero in the treatment-naive model (an estimated selection outcome, not an imposed constraint), so the paired analysis is reported as a change in program level and not as evidence that D_2 predicts survival.

28.1 Prognostic transportability of D1 in treated metastatic disease

To evaluate transportability in treated metastatic disease while retaining treatment-arm structure, we projected the frozen DeSurv programs into the O'Kane/COMPASS cohort and first estimated D_1 associations separately within each arm. Higher D_1 scores had protective point estimates in all three treatment arms: FOLFIRINOX (HR per SD 0.80, 95% CI 0.61–1.03), gemcitabine/nab-paclitaxel (HR 0.57, 95% CI 0.37–0.86) and the GA/experimental arm (HR 0.83, 95% CI 0.58–1.20; Fig. 4a). An arm-stratified Cox model allowing treatment-specific baseline hazards yielded a common association of HR 0.75, 95% CI 0.62–0.90 ($P = 0.003$; 173 patients, 123 deaths), with no evidence of heterogeneity across treatment arms (interaction $P = 0.40$). Because these specimens were epithelial-enriched by laser-capture microdissection, this analysis specifically tests transportability of the tumor-associated D_1 direction rather than the stromal programs.

28.2 Treatment-associated change in D2

In the 29 Linehan patients with paired pre- and post-treatment biopsies, the projected D_2 score increased significantly from pre- to post-treatment (paired Wilcoxon $P = 0.013$; mean $\Delta = +0.34$ SD; Fig. 4b), indicating a shift toward proCAF-associated bulk expression. This bulk analysis cannot distinguish a change in fibroblast state from altered tissue composition.

28.3 Caveats

These analyses are exploratory and do not establish treatment-predictive effects; treatment, stage and cohort are confounded, the treated cohorts are modestly sized, and no analysis plan was preregistered. The O'Kane/COMPASS analysis is prognostic within treated metastatic disease, whereas the Linehan analysis tests a paired change in program score. Each analysis is confined to a single cohort, and the

O'Kane/COMPASS common slope is estimated within that cohort from a model with arm-specific baseline hazards; it is not a pooled estimate across cohorts. We did not test a survival association for D_2 in the paired Linehan cohort; this analysis concerns program-level change. The trained survival coefficient was zero, but that alone does not preclude a separate association in another setting. A stromal D_2 readout is not interpretable in O'Kane/COMPASS, whose specimens are laser-capture epithelium rather than bulk tissue. Its D_1 result chiefly supports transport of the tumor-associated component of the score. Robust treated-cohort validation, and any assessment of whether the programs predict differential treatment benefit, will require larger harmonized series with adequately powered treatment-by-program interaction analyses.

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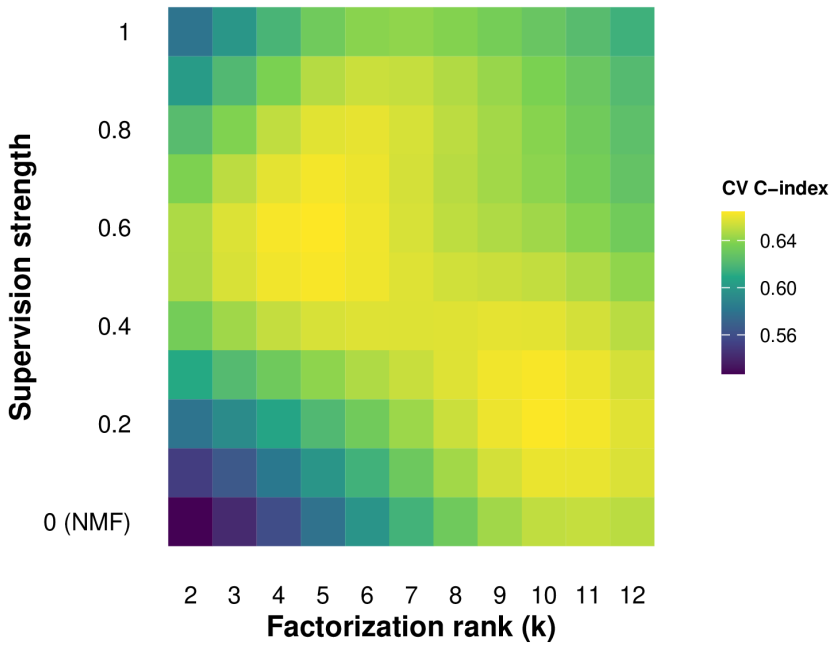


Figure S1: Bayesian-optimization tuning surface for DeSurv in the PDAC training cohort (TCGA + CPTAC). Gaussian-process–predicted mean cross-validated C-index over factorization rank (k) and supervision strength (α; $\alpha = 0$ corresponds to the unsupervised limit). The response surface shows a broad region of high cross-validated performance at intermediate supervision and moderate rank, a second high region at $k \geq 9$ and $\alpha \approx 0.2$–0.3, and discrimination degrading toward the unsupervised ($\alpha = 0$) boundary. The surface is a two-dimensional slice of the fitted five-parameter Gaussian process through the Bayesian-optimization-selected values of the top-gene count, ridge penalty and elastic-net mixing parameter (n_{top}, λ, ξ), not a marginal or profile surface over those hyperparameters.

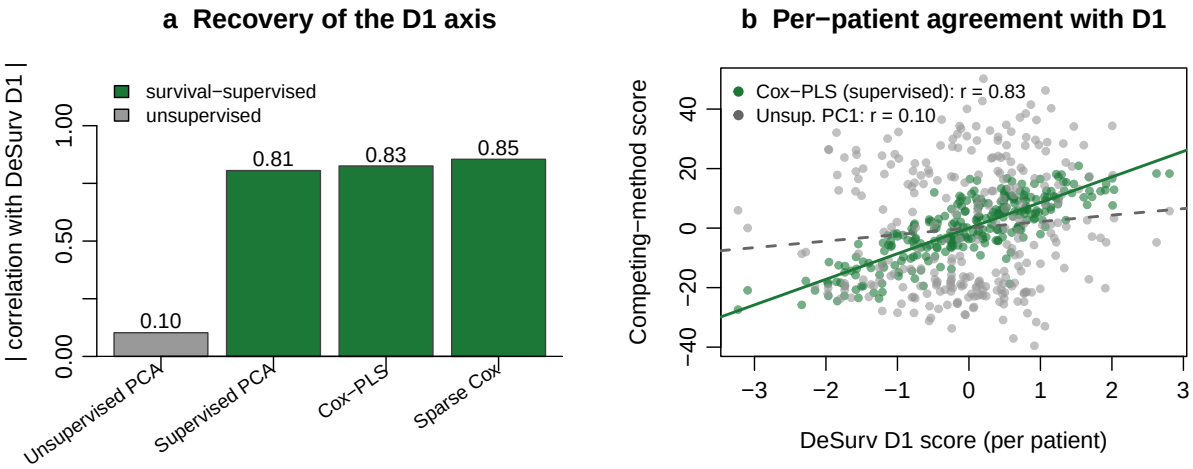


Figure S2: Correspondence of supervised scores and the first unsupervised principal component with D1 in the training cohort. (a) Absolute correlation (across training-cohort patients) of each method's leading direction with the DeSurv D1 score, in the training cohort: supervised principal components, Cox partial least squares, and sparse Cox are each strongly correlated with D1 (green), whereas the leading unsupervised principal component is only weakly correlated (grey). (b) Sample-level agreement with D1 for Cox-PLS (supervised, green) versus the first unsupervised principal component (grey).

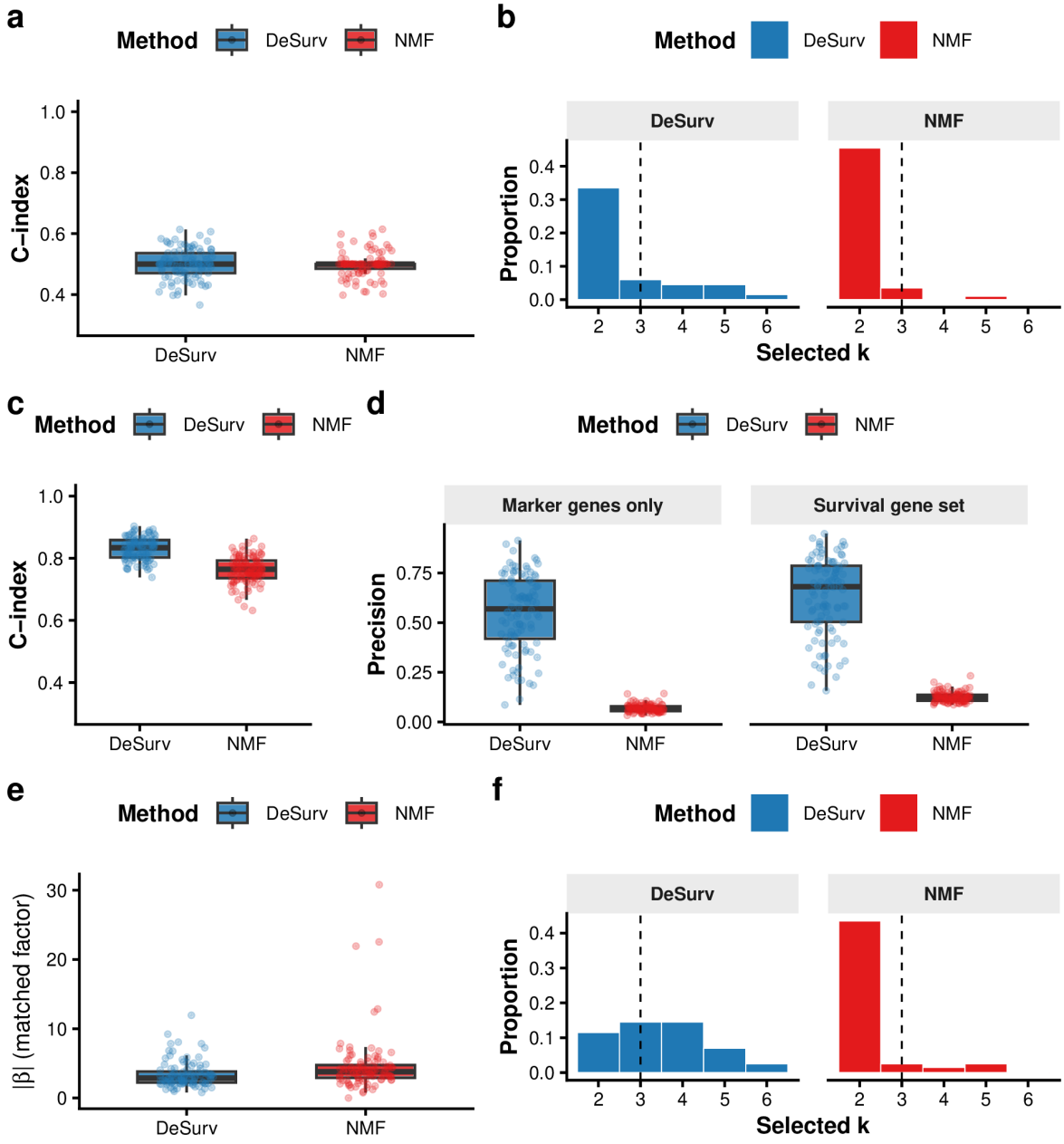


Figure S3: DeSurv performance under null and mixed simulation scenarios. (a,b) Null scenario ($\beta = 0$): both methods yield C-index ≈ 0.5 (a) and comparable rank distributions (b), so both methods show no out-of-sample prognostic discrimination under the null; precision is omitted as no true survival genes exist. (c–f) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (c) C-index is modestly higher for DeSurv. (d) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (e) The matched factor receives a larger fitted Cox coefficient $|\beta|$ under DeSurv. (f) Selected-rank distributions. DeSurv (blue); NMF (red, $\alpha = 0$); both Bayesian-optimized.

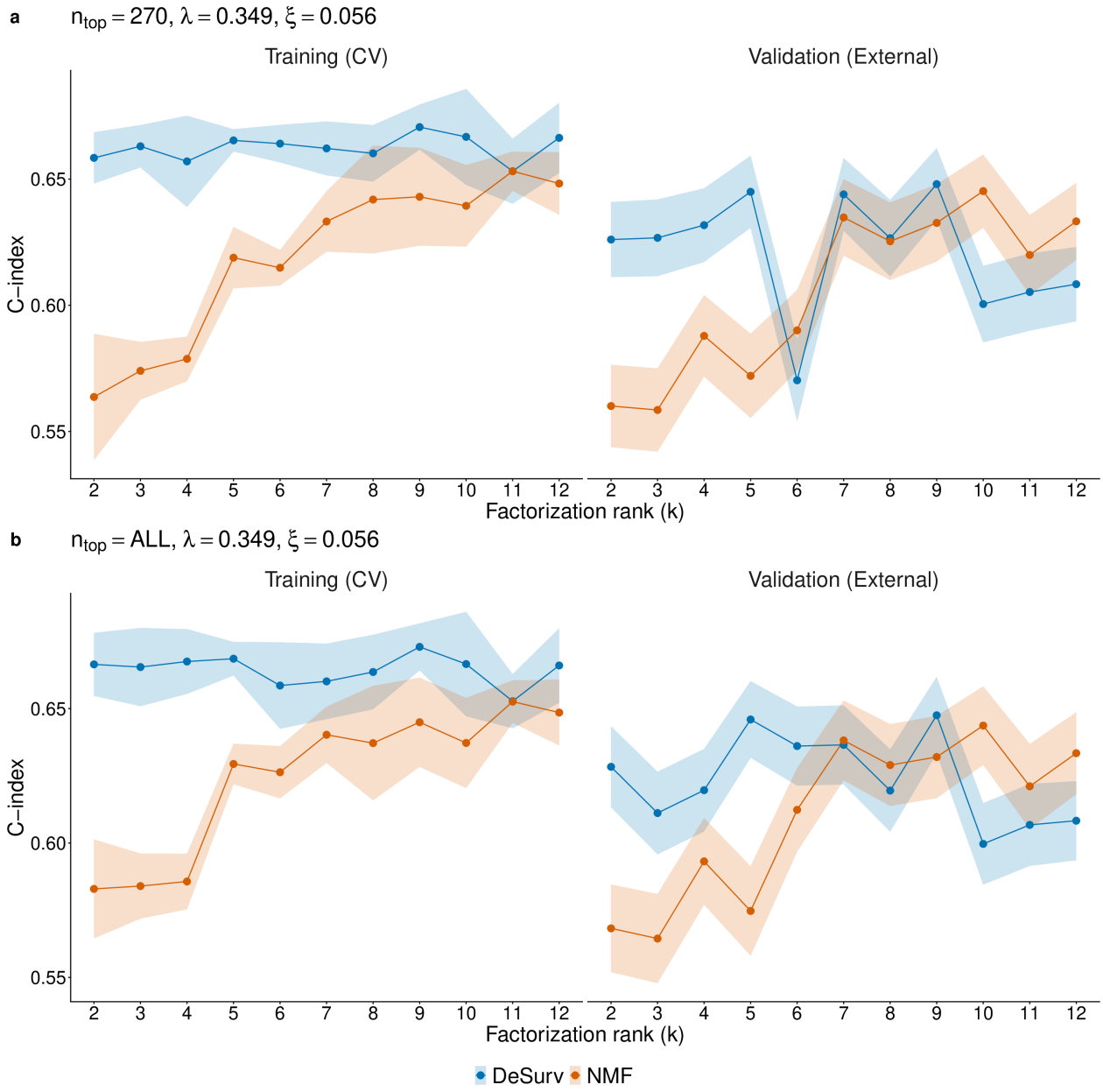


Figure S4: Training (CV) and external validation C-index as a function of factorization rank k for DeSurv (blue) and its unsupervised limit (red, $\alpha = 0$). Shaded ribbons show ± 1 SE. For each k, the best α was selected by maximizing the mean CV C-index. Validation C-index is a single Harrell concordance computed on the de-duplicated external patients of the four validation cohorts pooled together, so concordance pairs cross cohorts, with the ribbon giving plus or minus one standard error from that single concordance estimate. This descriptive quantity differs from the primary combined external C-index reported in the main text, which is an event-weighted mean of four cohort-specific C-indices, and the two are not directly comparable. (a) Results using $n_{\text{top}} = 270$ top genes per factor ($\lambda = 0.349$, $\xi = 0.056$). (b) Results retaining all genes ($n_{\text{top}} = \text{ALL}$; $\lambda = 0.349$, $\xi = 0.056$).

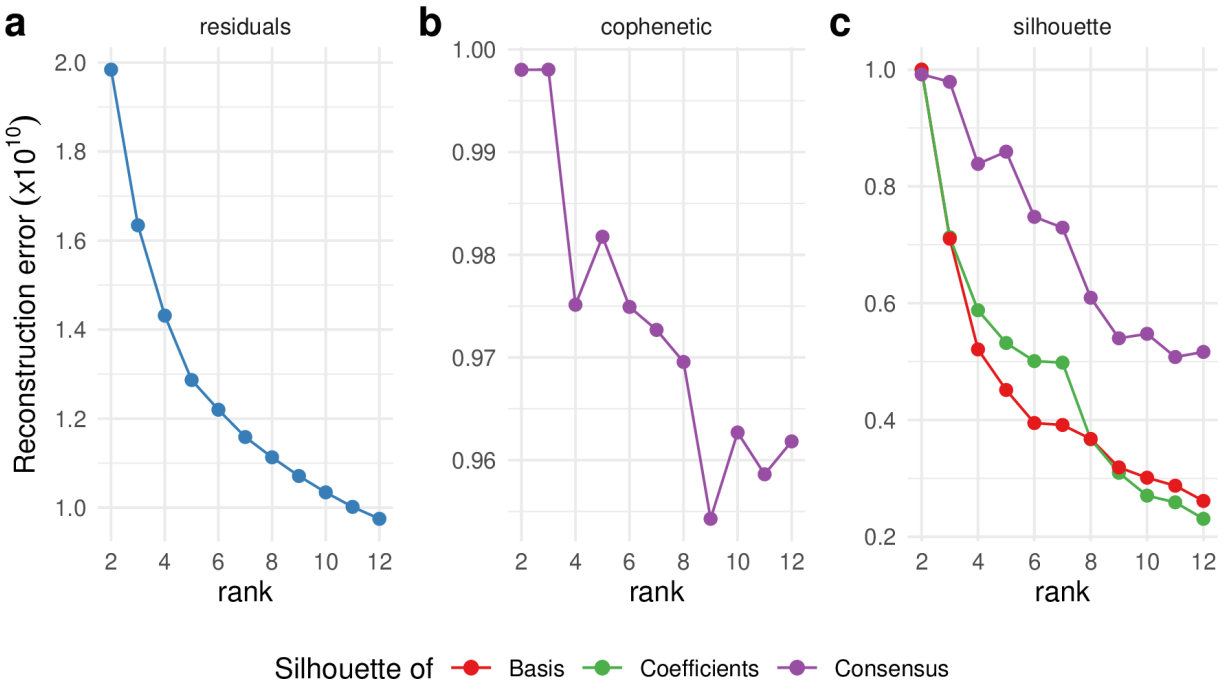


Figure S5: Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (a) Reconstruction residuals as a function of factorization rank k. (b) Cophenetic correlation coefficient as a function of k. (c) Mean silhouette width of the basis, coefficient and consensus matrices as a function of k. These criteria yield inconsistent guidance, illustrating the ambiguity of rank selection under unsupervised criteria; the main text selects rank on cross-validated prognostic performance.

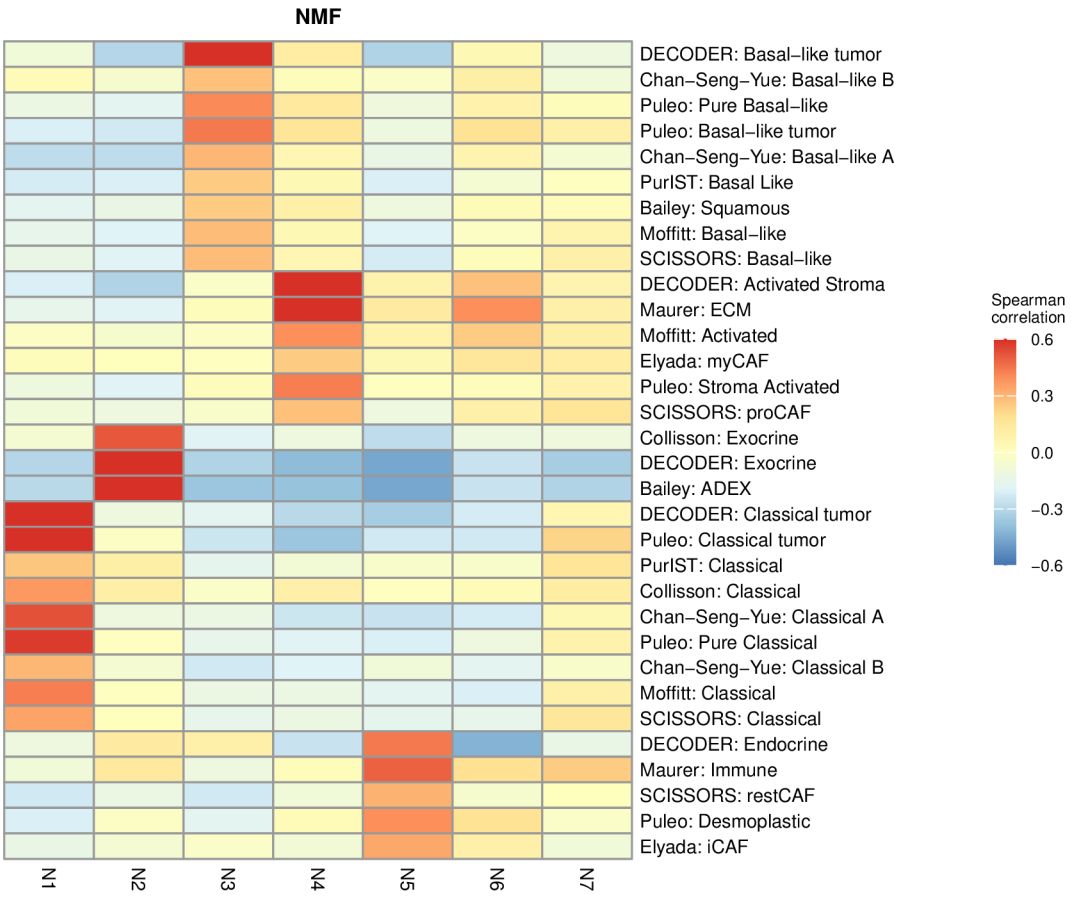


Figure S6: Gene-program correspondence for the rank-optimized unsupervised control ($k = 7$, BO-selected rank at $\alpha = 0$), computed with the same Spearman-correlation analysis and reference programs as main-text Fig. 2a,b (top 50 genes per factor; reference programs with $ho > 0.2$ for at least one factor are shown). Compared with the matched-rank $k = 3$ solution, the $k = 7$ solution shows broader correspondence to established PDAC programs, with some programs represented by more than one factor. Supplementary Fig. S7 quantifies the accompanying changes in coverage and multiplicity.

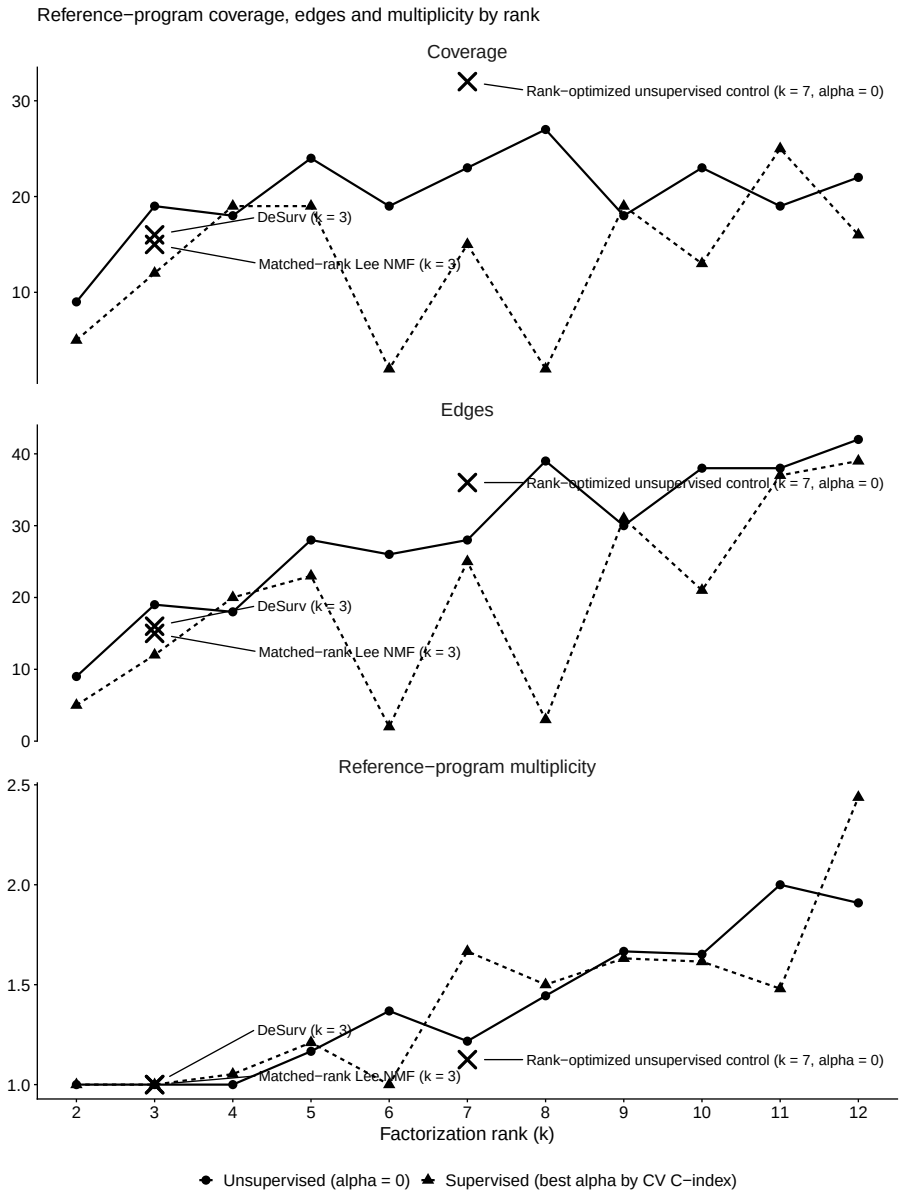


Figure S7: Reference-program coverage, edges and multiplicity by factorization rank k, for the unsupervised ($\alpha = 0$) and supervised (best α by cross-validated C-index) curves, computed from the same factor-to-reference-program correspondence analysis as Fig. S6 and main-text Fig. 2a,b (top 50 genes per factor, $ho > 0.2$ threshold, same reference-program universe). Rank-wise points are single fits at fixed λ, ξ and n_{top}, not the 100-restart consensus fits used elsewhere; markers denote the three final-model anchors (the consensus fits for DeSurv $k = 3$ and the rank-optimized unsupervised control $k = 7$, and the best-of-30-run `NMF::nmf` fit at matched rank $k = 3$).

Table S1: Where survival supervision enters, and what a new sample requires, for DeSurv and five comparators. Cells for DeSurv, standard NMF, supervised principal components and penalized Cox regression are taken from the Methods of this manuscript; CoxNMF cells from its published objective function and SurvNMF cells from its full text; both works are cited in the text of this section. "Not described in the source" means the cited work does not state the behavior; no other cell is inferred. The table records architecture and does not rank the methods. Note: CoxNMF does not describe a procedure for scoring new samples. The implication that sample-side loadings would need to be obtained for new samples follows from its published objective and is an inference, not a reported implementation.

Method	Learned representation	Where survival enters	Gene-program basis directly supervised?	New-sample scoring	External latent-variable re-estimation?
Standard NMF	Gene programs W and sample loadings H	Not in the factorization; Cox fitted afterwards on projected scores (Methods)	No	In this study, projection onto the fixed W, as for DeSurv (Methods)	No (projection)
DeSurv (this work)	Gene programs W, sample loadings H and Cox coefficients β on $Z = W^\top X$	Cox partial likelihood on $Z = W^\top X$; its gradient acts on W (Methods)	Yes	Projection $Z = \widetilde{W}^\top X_{\text{new}}$ with $\widetilde{W}$ frozen (Methods)	No
CoxNMF	Basis W (genes) and coefficients H (samples), with Cox coefficients β on H	Cox partial likelihood whose covariates are the columns of H (published objective)	No: the Cox term does not depend on W (published objective)	Not described in the source	Not described in the source
SurvNMF	Sample-side latent matrix and feature matrix (the former is denoted W in SurvNMF's transposed notation; full text)	Cox model fitted jointly on the sample-side latent matrix (full text)	No: the Cox term is on the sample side (full text)	Held-out loadings re-solved by constrained least squares (HALS) with the feature matrix fixed (full text)	Yes, for held-out samples; reported for simulated data only (full text)
Supervised principal components	Principal components of genes passing a univariate Cox score threshold	Gene screening by Cox score before the components are formed (Methods)	No: components are unsupervised on the screened genes	Fixed linear map over the screened genes (Methods)	No
Penalized Cox regression	One coefficient vector over genes (a single linear predictor)	Directly: the objective is the penalized Cox likelihood (Methods)	No gene programs are learned	Fixed linear predictor (Methods)	No

Table S2: Cross-validation of the final DeSurv hyperparameter configuration. The overall-best $k = 7$ configuration, the best $k = 3$ configuration identified during Bayesian optimization, and the final fitted $k = 3$ configuration were re-evaluated using the same DeSurv software build and the Bayesian-optimization cross-validation protocol: five folds constructed within the training data and stratified jointly by event status and cohort (seed 123), 30 random initializations per configuration and fold, and the same preprocessing, stopping tolerance, iteration cap and n_{top}-based scoring procedure. Values are mean Harrell C-index across folds with fold-level standard error. The cached Bayesian-optimization history was generated with an earlier DeSurv installation whose package version was not recorded; the best $k = 3$ configuration has C-index 0.647 under the current build versus 0.646 in the historical optimization object. The same-build values shown here are therefore used for direct numerical comparison. The final configuration lies 0.002 below the recomputed one-standard-error threshold of 0.644 (0.655 minus its fold-level standard error).

Configuration	k	α	λ	ξ	n_{top}	Mean C-index	Fold-level SE
Overall best ($k = 7$)	7	0.334	0.349	0.056	270	0.655	0.011
Best $k = 3$ in the search	3	0.442	0.241	0.201	299	0.647	0.009
Final fitted model ($k = 3$)	3	0.334	0.349	0.056	270	0.642	0.007

Table S3: Summary of PDAC cohorts used in this study. N denotes the number of samples after all quality-control filters (histology, survival validity, dataset-specific exclusions). Patient-level metadata: OS = overall survival (time and event indicator); PurIST = tumor subtype classification; DeCAF = CAF subtype classification. Expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study. Events denotes the number of overall-survival events (deaths) with the corresponding event rate; outcome frequency was lower and more homogeneous in the training cohorts (50–52%) than across the validation cohorts (60–90%, pooled 67%), a case-mix difference relevant to interpreting external-validation performance.

Cohort	Platform	N	Events	Role	Accession	Metadata	Reference
TCGA-PAAD	RNA-seq	144	75 (52%)	Training	GDC (TCGA-PAAD)	OS, PurIST, DeCAF	Raphael et al. (2017)
CPTAC	RNA-seq	129	64 (50%)	Training	GDC (CPTAC-3)	OS, PurIST, DeCAF	Cao et al. (2021)
Dijk	RNA-seq	90	81 (90%)	Validation	E-MTAB-6830	OS, PurIST, DeCAF	Dijk et al. (2020)
Moffitt	Microarray	123	83 (67%)	Validation	GSE71729	OS, PurIST, DeCAF	Moffitt et al. (2015)
PACA-AU (array)	Microarray	63	38 (60%)	Validation	GSE36924	OS, PurIST, DeCAF	Bailey et al. (2016)
PACA-AU (RNA-seq)	RNA-seq	52	31 (60%)	Validation	EGAS00001000154	OS, PurIST, DeCAF	Bailey et al. (2016)
Puleo	Microarray	288	181 (63%)	Validation	E-MTAB-6134	OS, PurIST, DeCAF	Puleo et al. (2018)
Training total		273	139 (51%)				
Validation platform profiles total		616	414 (67%)				
Combined unique-patient validation set		570	388 (68%)				

PACA-AU contains 46 patients profiled on both the array and RNA-seq platforms; combined (pooled) analyses represent each patient once, retaining the RNA-seq profile for duplicated patients, consistent with the DeCAF preprocessing rule. Platform-specific per-dataset analyses retain both records.

Table S4: The 45-program reference universe used for every factor-to-program correspondence analysis (main-text Fig. 2a,b; Supplementary Figs. S6 and S7), generated from the same renamed and de-duplicated object the analyses read. Displayed alias is the row label used in the figures; Genes is the size of the published list; In analysis set is the number of those genes among the 1,970 analyzed genes, over which the Spearman correlation is computed. The SCISSORS restCAF and proCAF rows are the SCISSORS iCAF and myCAF signatures under DeCAF nomenclature. The two Puleo basal-like rows are different programs: the whole-tumor pure basal-like component and the tumor-compartment basal-like signature.

Displayed alias	Source	Genes	In analysis set
PurIST: Basal Like	Rashid et al. 2020	8	7
PurIST: Classical	Rashid et al. 2020	8	8
DECODER: Basal-like tumor	Peng et al. 2019	394	139
DECODER: Classical tumor	Peng et al. 2019	372	166
DECODER: Activated Stroma	Peng et al. 2019	206	108
DECODER: Normal Stroma	Peng et al. 2019	97	15
DECODER: Immune	Peng et al. 2019	408	60
DECODER: Exocrine	Peng et al. 2019	113	44
DECODER: Endocrine	Peng et al. 2019	380	61
Moffitt: Basal-like	Moffitt et al. 2015	25	20
Moffitt: Classical	Moffitt et al. 2015	25	21
Moffitt: Activated	Moffitt et al. 2015	25	24
Moffitt: Normal	Moffitt et al. 2015	25	11
Moffitt: Immune	Moffitt et al. 2015	25	5
Collisson: Classical	Collisson et al. 2011	22	22
Collisson: Exocrine	Collisson et al. 2011	20	19
Collisson: QM	Collisson et al. 2011	20	12
Bailey: Squamous	Bailey et al. 2016	87	27
Bailey: Immunogenic	Bailey et al. 2016	145	22
Bailey: Pancreatic Progenitor	Bailey et al. 2016	73	7
Bailey: ADEX	Bailey et al. 2016	57	37
Chan-Seng-Yue: Basal-like A	Chan-Seng-Yue et al. 2020	100	34
Chan-Seng-Yue: Basal-like B	Chan-Seng-Yue et al. 2020	100	28
Chan-Seng-Yue: Classical A	Chan-Seng-Yue et al. 2020	100	69
Chan-Seng-Yue: Classical B	Chan-Seng-Yue et al. 2020	100	31
Maurer: ECM	Maurer et al. 2019	222	139
Maurer: Immune	Maurer et al. 2019	251	101
Puleo: Pure Classical	Puleo et al. 2018	90	71
Puleo: Immune Classical	Puleo et al. 2018	56	7
Puleo: Desmoplastic	Puleo et al. 2018	101	65
Puleo: Stroma Activated	Puleo et al. 2018	62	50
Puleo: Pure Basal-like	Puleo et al. 2018	94	57
Puleo: Basal-like tumor	Puleo et al. 2018	461	118
Puleo: Classical tumor	Puleo et al. 2018	539	273
Elyada: iCAF	Elyada et al. 2019	35	25
Elyada: myCAF	Elyada et al. 2019	16	15
SCISSORS: apCAF	Leary et al. 2023	25	4
SCISSORS: restCAF	Leary et al. 2023	25	17
SCISSORS: proCAF	Leary et al. 2023	25	16
SCISSORS: Perivascular	Leary et al. 2023	25	5
SCISSORS: panCAF	Leary et al. 2023	25	22
SCISSORS: Basal-like	Leary et al. 2023	25	24
SCISSORS: Classical	Leary et al. 2023	25	22
deCAF: proCAF	Peng et al. 2026	9	4
deCAF: restCAF	Peng et al. 2026	9	8

Table S5: Index of analysis scripts. Scripts are numbered as in the code repository; helper functions shared by several scripts (de-duplication of PACA-AU patients, validation scoring, elbow selection) are called from the scripts listed. Section numbers refer to this document.

Script	Analysis	Read in
code/03	Bayesian optimization of hyperparameters; the matched-budget refinement with α pinned at 0	Sections 5, 26.3
code/04	Final fits: 100-restart consensus fits for DeSurv and the two $\alpha = 0$ controls; best-of-30-run `NMF::nmf` fits for standard NMF at matched and elbow rank, each with a post hoc Cox fit	Sections 5.1, 21
code/05	External validation: frozen projection, linear predictor and per-dataset C-index	Sections 7, 17
code/06	Fixed-hyperparameter $k \times \alpha$ cross-validation grid and its external-validation curve	Sections 24, 26.1
code/07	Simulation studies (three scenarios, 100 replicates each)	Section 23
code/11	Treated-cohort analyses (O'Kane/COMPASS arm-stratified D1; Linehan paired D2)	Section 28
code/14	Variance partition of the D1 score; additive and interaction adjustment for PurIST and DeCAF	Section 18
code/15	Single-component supervised comparators and recovery of the D1 axis	Section 13
code/16	Subsample stability of top-gene sets across methods	Section 21
code/17	Proper-scoring sensitivity (integrated Brier score, IPA)	Section 15
code/18	Clinical-covariate and tumor-purity adjustment by cohort	Section 19
code/19	Independently tuned supervised comparators, axis decomposition and compartment attribution	Section 14
code/20	Dispersion of individual random restarts	Section 21
code/21	Reproducibility of the consensus endpoint from disjoint restart halves	Section 21
code/23	Stromal reference-program resolution (top-gene and full-gene views)	Section 26.2
code/24	Reference-program coverage, edges and multiplicity by rank; the reference-universe table	Sections 8, 26.1
code/25	Between-cohort heterogeneity, proportional-hazards diagnostic and bootstrap C-index intervals	Sections 16, 17
code/26	Reconstruction of the gradient-scaling factor ζ along the production optimization path	Sections 2.3, 21
code/29	Re-evaluates the overall-best, best $k = 3$ and final hyperparameter configurations under the Bayesian-optimization cross-validation protocol (Table S2)	Section 5

Table S6: Cross-method projection R^2: how reconstructable each factor's gene-loading vector is from the other method's full basis (OLS R^2). DeSurv retains the bulk of the unsupervised structure (NMF N1 and N3 carry over at $R^2 \geq 0.88$) and reorganizes it: the survival-aligned D1 program is essentially unreconstructable from the NMF basis ($R^2 = 0.09$), while the exocrine NMF factor N2 is poorly reconstructed from the DeSurv basis ($R^2 = 0.38$).

Factor	R^2 from the other method's basis
NMF N1 (tumor)	0.90
NMF N2 (exocrine)	0.38
NMF N3 (microenviron.)	0.88
DeSurv D1 (classical/restCAF-like)	0.09
DeSurv D2 (proCAF-like)	0.81
DeSurv D3 (basal-like)	0.75

Table S7: Whole-matrix reconstruction R^2 across the 1,970-gene analysis matrix for standard NMF versus DeSurv at $k = 3$. Two denominators are shown. The centered R^2 is the conventional definition, taken about the grand mean, and is the one to read as fraction of variation reconstructed. The uncentered R^2 divides by $\sum X^2$; because the analysis matrix holds within-sample ranks with grand mean $p/2$, the mean alone accounts for about 75% of that denominator, so uncentered values are high for any nonnegative low-rank fit and are reported for completeness. On the centered definition DeSurv gives up 6.0 percentage points of reconstruction relative to NMF (1.5 on the uncentered scale).

Model	Centered R^2	Uncentered R^2	Residual fraction (centered)
NMF ($k = 3$)	0.812	0.953	0.188
DeSurv ($k = 3$)	0.752	0.938	0.248

Table S8: Leave-one-dataset-out sensitivity of the pooled external-validation hazard ratio for the frozen DeSurv linear predictor. Each row refits the dataset-stratified Cox model on the de-duplicated patient-level table with the named validation dataset omitted; the first row is the full combined estimate reported in main-text Table 1. Hazard ratios are per pooled standard deviation of the linear predictor, recomputed within each subset, the same convention as the headline pooled estimate and not the within-cohort convention of Table S11. Puleo contributes 288 of the 570 unique patients, which is why this sensitivity analysis is reported.

Dataset omitted	n	Deaths	HR per pooled SD (95% CI)	P
None (all datasets)	570	388	1.42 (1.26–1.59)	$< 10^{-4}$
Dijk	480	307	1.45 (1.27–1.67)	$< 10^{-4}$
Moffitt GEO array	447	305	1.44 (1.27–1.64)	$< 10^{-4}$
PACA-AU array	553	376	1.41 (1.26–1.59)	$< 10^{-4}$
PACA-AU RNA-seq	518	357	1.38 (1.22–1.56)	$< 10^{-4}$
Puleo	282	207	1.34 (1.17–1.55)	$< 10^{-4}$

Table S9: Hazard ratios (per SD of the DeSurv linear predictor) for overall survival, before and after adjustment for standard clinical covariates and, where annotated, tumor purity, fitted separately in each cohort by complete-case analysis (per-cell n and events shown). The Covariates column lists the clinical variables entered (stage, grade, age, sex, nodal status, or resection margin, as available). After clinical adjustment the DeSurv score remains independently prognostic in Puleo, PACA-AU array and both training cohorts, and is attenuated to non-significance in PACA-AU RNA-seq, Dijk and Moffitt, the three smallest complete-case sets. Where tumor-purity annotation was available, purity adjustment left the estimate essentially unchanged wherever it could be measured on the full cohort (PACA-AU array 1.44 versus 1.45, PACA-AU RNA-seq 1.66 versus 1.65, TCGA 3.50 versus 3.60; unadjusted versus purity-adjusted), providing no evidence that the association was explained by measured tumor purity. The smaller Dijk and Moffitt cohorts are individually underpowered, and their estimates are correspondingly unstable: Moffitt is non-significant both unadjusted and after adjustment, and its purity-adjusted model rests on only $n = 29$ of its 123 patients. Training-cohort (TCGA, CPTAC) hazard ratios are in-sample.

Cohort	Covariates	Unadjusted	Clinical-adjusted	Purity-adjusted
PACA-AU array	stage, grade, age, sex	1.44 (1.07–1.93), 0.016 (n=63, 38 ev)	1.58 (1.06–2.35), 0.025 (n=62, 37 ev)	1.45 (1.08–1.96), 0.014 (n=63, 38 ev)
PACA-AU RNA-seq	stage, grade, age, sex	1.66 (1.15–2.38), 0.006 (n=52, 31 ev)	1.56 (0.95–2.55), 0.077 (n=51, 30 ev)	1.65 (1.15–2.36), 0.006 (n=52, 31 ev)
Puleo	sex, margin	1.37 (1.18–1.59), $<$ 0.001 (n=288, 181 ev)	1.33 (1.14–1.55), $<$ 0.001 (n=284, 177 ev)	—
Dijk	grade, age, sex, nodal	1.32 (1.06–1.66), 0.015 (n=90, 81 ev)	1.18 (0.92–1.50), 0.189 (n=85, 77 ev)	—
Moffitt	sex, margin	1.20 (0.98–1.48), 0.076 (n=123, 83 ev)	1.20 (0.94–1.53), 0.154 (n=98, 67 ev)	0.87 (0.50–1.54), 0.639 (n=29, 18 ev)
TCGA (train)	stage, grade, age, sex, nodal	3.50 (2.63–4.66), $<$ 0.001 (n=144, 75 ev)	3.32 (2.40–4.59), $<$ 0.001 (n=142, 74 ev)	3.60 (2.68–4.84), $<$ 0.001 (n=143, 74 ev)
CPTAC (train)	stage, age, sex, nodal	3.72 (2.73–5.08), $<$ 0.001 (n=129, 64 ev)	3.59 (2.59–4.97), $<$ 0.001 (n=123, 61 ev)	—

Table S10: External validation C-index by method and validation dataset. DeSurv at the one-standard-error-selected rank ($k = 3$) is compared with standard NMF at matched rank ($k = 3$) and at elbow-selected rank ($k = 5$), with the matched-budget unsupervised control ($k = 3$, $\alpha = 0$) and with the rank-optimized unsupervised control ($k = 7$, $\alpha = 0$). The unsupervised columns are not all produced by one implementation: the two standard NMF columns are `NMF::nmf`, whereas the two controls are the DeSurv optimizer run at $\alpha = 0$ with the same consensus-initialization procedure as the supervised model. The matched-budget column is the tuning-budget control: it holds rank, optimizer, consensus-initialization procedure and Bayesian-optimization budget fixed while re-optimizing the remaining hyperparameters with $\alpha = 0$. Its external discrimination is no better than that of `NMF::nmf` at the same rank, a comparison made across the two implementations described here. Higher values indicate better discrimination. Every method was scored on the union of its own factors' top-n_{top} gene supports at the supervised model's $n_{\text{top}} = 270$, the same construction as the pooled hazard ratios, main-text Table 1 and Table S11. Scoring the matched-budget control at its own tuned $n_{\text{top}} = 299$ instead changes no per-dataset C-index by more than 0.007; the rank-optimized control was likewise scored at 270 rather than at its own tuned $n_{\text{top}} = 187$.

Dataset	DeSurv ($k = 3$)	standard NMF ($k = 3$)	Matched-budget control ($k = 3$, $\alpha = 0$)	standard NMF ($k = 5$, elbow)	Rank-optimized control ($k = 7$, $\alpha = 0$)
Dijk	0.607	0.542	0.516	0.633	0.631
Moffitt	0.562	0.521	0.516	0.552	0.578
PACA-AU array	0.649	0.470	0.462	0.657	0.669
PACA-AU RNA-seq	0.634	0.430	0.412	0.679	0.702
Puleo	0.636	0.534	0.533	0.642	0.651

Table S11: Cohort-specific external-validation estimates for the frozen DeSurv linear predictor, on the de-duplicated patient-level table (570 unique patients, 388 deaths; PACA-AU represented once, RNA-seq retained). Hazard ratios are per standard deviation of the linear predictor computed within that cohort, the convention used for the interaction test, so they are not comparable to the pooled-standard-deviation hazard ratio in main-text Table 1. C-index confidence intervals are percentile intervals from 1000 stratified bootstrap replicates (seed 1) resampling within cohort; the combined C-index is the event-weighted mean of the four cohort values.

Cohort	n	Deaths	HR per within-cohort SD (95% CI)	C-index (95% CI)
Dijk	90	81	1.32 (1.06–1.66)	0.607 (0.531–0.679)
Moffitt	123	83	1.20 (0.98–1.48)	0.562 (0.478–0.639)
PACA-AU	69	43	1.46 (1.09–1.96)	0.630 (0.541–0.745)
Puleo	288	181	1.37 (1.18–1.59)	0.636 (0.594–0.678)
Combined	570	388	—	0.613 (0.579–0.646)

Table S12: Top 270 genes for DeSurv factor D1, classical tumor and restCAF-like stroma, ranked by factor specificity score s_{ij}. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension (n_{top}).

Factor D1 (classical tumor + restCAF-like stroma)					
1–45	46–90	91–135	136–180	181–225	226–270
DMBT1	FGFR2	GOLGA8A	PTPRN2	RPL9	PPP1R12B
IGFBP2	COL9A2	IL2RG	FCGBP	CD79A	PIP5K1B
S100A4	COL22A1	KIAA1211	PLA2G2A	PROM1	AHCYL2
KRT23	IRF7	SFRP1	FABP4	SCD	SST
LIF	SBNO2	ADH1B	EPPK1	TMC5	CYP3A5
IFI6	MGAT3	MFSD2A	CPXM2	CCL14	SMAP2
PDLIM3	SYNM	CLU	KIF1A	PLXNB1	ZG16B
CLDN18	APOD	CLMN	NGFR	TMEM119	SERPINF1
MYO15B	SEMA6A	CHPF	NFKBIA	SORBS2	CLDN3
SYT8	TNXB	MTMR11	SPOCK2	C7	NR1I2
TFF2	PTGIS	TTR	ATP2C2	SLC12A7	JUND
TMPRSS2	CCL18	WFDC2	NBEAL2	QSOX1	NCF2
CHGB	ZC3H12A	C2orf72	CAPN9	UNC5CL	CELSR1
CST1	CELSR3	CCND2	PTGDS	LONRF2	MUC17
B3GNT7	EMB	ARHGAP4	LSP1	VSIG2	INPP5D
F5	KCNJ15	PKDCC	LAMA5	ADAMTSL2	TM4SF5
TNRC18	SORBS1	SRPX	FASN	ARHGEF16	HNF4A
IGF2	PAQR8	CCL21	GFRA1	PCOLCE	CFD
ARSD	SLC4A4	CARD11	PTPRH	CFC1B	HCLS1
ZDHHC7	ACSS1	CTRB2	FMOD	SLC9A3	SH3PXD2B
LENG8	ST6GALNAC1	GOLGA8B	FHL1	ACACB	PCSK2
ATP9A	DES	IYD	PCDH20	BCAS1	HMGCS2
TMPRSS3	PRUNE2	CORO1A	CCR7	RASD1	FILIP1L
CES1	PLCB4	SAA2	CHIT1	LGALS4	ETNK1
HOPX	CADPS	CDCA7	PARM1	MS4A1	TNNI2
CNN1	FMO5	BOC	ITM2A	CAPN6	TOX3
MIA	RAP1GAP2	LTB	NAPSB	SCCPDH	FRZB
CRIP2	GALNT12	COMP	AGT	PTPRCAP	HABP2
NEURL1B	CCL2	MANSC1	FUT3	VNN1	GDA
CHGA	REG3A	SLC19A3	SIPA1L2	PI16	NCOR2
ACTG2	ZCCHC24	STEAP2	SPINK5	SEMA7A	DNAH2
CXCL14	LTF	FOXA1	ACSL1	IL10RA	DOK4
TPPP3	EFHD2	WNK2	BTNL8	GALNT3	HOXB9
PIGR	ATP11A	SELL	CORO2A	GOLM1	KIAA1324
CYP27A1	GP2	PLEKHA6	LAD1	TNFAIP2	TRIM2
ADAM8	HIPK2	NFASC	TMEM63A	CD79B	FGFBP1
FGFR3	SLC26A9	EGR3	GC	ENPP2	IRX3
IGF1R	TACC2	PLIN4	C1orf21	SQLE	TINAGL1
PLEKHB1	TRAK1	WDR72	MIF	KLF4	SFRP4
CPXM1	SAA1	AMY2A	FOXA2	PDLIM7	APLNR
ATP7B	CLDN23	IL16	CCL19	FOSB	AGRN
SLC12A2	FOXJ1	TJP3	GSTA1	RHPN2	SLC25A25
CCL20	FRY	SLC9A2	LLGL2	SCG2	PODN
AKNA	GATA6	LMOD1	ACP5	EMILIN1	ITIH2
COL16A1	PLXNA2	CPS1	UGT2B15	BACE2	FLJ23867

Table S13: Top 270 genes for DeSurv factor D2, proCAF-like stroma, ranked by factor specificity score s_{ij}.

Factor D2 (proCAF-like stroma)					
1–45	46–90	91–135	136–180	181–225	226–270
UHMK1	RPS26	SAMD9L	ATP8A1	IRF6	RASSF2
DDR2	MYLK	DOCK2	TGFBR1	KLHL6	ZNF83
INHBA	COL11A1	IGF1	SYNPO2	C1QTNF3	CYTIP
DYNC2H1	NFIB	TACC1	PDE4B	PCLO	SRPX2
ZDHHC20	PTPN13	CFH	SGCD	ITGA2	LY75
ITPR2	PDE3A	DMD	STK17B	LRIG3	HLA-DOA
HMCN1	TRIM22	PCDH7	NUAK1	SSPN	STK38L
LAMA2	DPYD	MALAT1	FCGR3A	F2R	RSAD2
SLFN5	COL14A1	NOTCH2	GPNMB	PDGFC	HBB
PLXNC1	CNTN1	CDK6	VCAM1	CD109	HERC2P2
SVEP1	NCKAP1L	FBN1	ADAMTS9	CLIC4	AOX1
CYBB	PHLDB2	SYNE2	ITGB8	SORL1	PLIN2
SYNE1	ITGA4	RHOBTB3	CR1	AOAH	PFKFB2
SAMHD1	MACC1	CALCRL	SEPT11	FMO2	OGN
SKIL	SEMA5A	NRP1	IFI44L	RDX	DOCK10
ADAMTS12	MRC1	LIFR	ELL2	IL1RAP	RGS4
BICC1	SLIT2	STAB1	GPR183	GBP1	AXL
ITGA1	ZEB2	FLRT2	ZBTB16	CXCR4	FGD6
COL8A1	ADAM10	RNF144A	FRMD6	WNK1	CDC42BPA
PDGFRA	SAMD9	BCAT1	PDE5A	MFAP5	SLC6A6
ABCA1	MS4A7	CILP	CSF1R	ALDH1A3	RGS1
TCF4	EFEMP1	CCDC80	PLEK	CD53	KIAA1324L
CEP170	BIRC3	FAT4	SLK	OMD	CPA3
CD163	WISP1	BNC2	SLC7A2	PLOD2	CHL1
MAN1A1	LAMA4	LTBP1	HEG1	ITGAV	SLC7A11
MSR1	FAM198B	COL12A1	GREM1	MACF1	GFPT2
CCDC88A	PTPRC	IL1R1	VGLL3	NID2	RAI14
F13A1	FAP	SLA	GBP5	NAMPT	LMAN1
RPS28	ARRDC3	MBOAT2	SLC1A3	ADAM28	MECOM
RALGAPA2	MYH10	FNDC3B	FGF7	GNE	FPR1
ZEB1	FCGR2A	PRRX1	ABCA8	ENAH	EDEM3
FBXO32	PARP14	COL10A1	ADAM12	FAM129A	RASSF6
INPP4B	DSE	DDX60	CYBRD1	ECM2	IQGAP2
CDH11	EHF	DOCK9	TFRC	VSIG4	MIAT
PKHD1	OSMR	SRGN	CTSK	MTAP	PRICKLE1
AKAP2	ARID5B	FOXO1	COLEC12	INMT	CHST15
CYP1B1	ASPN	MS4A6A	ALDH1L2	SGMS2	GEM
CCL5	SLCO2B1	ADAMTS2	PGM2L1	COL4A4	CHRDL1
FGL2	FPR3	CPVL	FAM135A	CAPRIN2	DSC2
MAP1B	DOCK8	CD93	DOCK11	CENPF	RAB31
PLXDC2	SLFN11	DST	EPHA3	DCLK1	SLC38A1
LRRK2	ZNF532	CDK14	PAPPA	EDNRA	CNN3
KIAA0754	SPON1	MSRB3	EGFR	WNT5A	FLT1
EDIL3	SEMA3C	RNF213	RASA1	GALNT5	TMEM200A
ITGBL1	FERMT2	SEL1L	AHR	SLC38A2	SLC2A3

Table S14: Top 270 genes for DeSurv factor D3, basal-like tumor, ranked by factor specificity score s_{ij}.

Factor D3 (basal-like tumor)					
1–45	46–90	91–135	136–180	181–225	226–270
HSPB1	HLA-H	LAMB3	MYL9	IL1RN	SDC1
KRT7	HSPA1A	FSCN1	COL17A1	TST	KLK11
MSLN	LGALS3	TSPAN13	EPS8L1	RAPGEFL1	PERP
RPSAP58	FAM83H	TSPAN8	CDH3	C15orf48	ARHGAP23
S100P	EPB41L1	ALDH3B1	RNASET2	PFKP	HPGD
RPS16	EFNA1	ANPEP	TUBA1C	STARD10	AKR1C3
IL32	TKT	GSTP1	RAP1GAP	FOXQ1	CRABP2
SH3BGRL3	SLC2A1	KLK10	H19	REG4	GRB7
IFITM2	PLOD1	B4GALT1	LTBP3	GJB1	SERPINA4
RPS21	IER3	GAS6	AQP3	SLC29A1	SLC39A4
SEMA4B	B3GNT3	ABCC3	MPZL2	NOTCH3	FA2H
APOL1	PRSS3	CLDN2	ID1	TRIM16	RPL8
S100A16	TOB1	CEACAM6	RARRES2	TAP2	MT1X
DNAJB1	EGR1	CREB3L1	SLC25A6	KRT19	TCN1
MT2A	PLA2G16	MST1R	RBP1	CRP	AKAP1
TNFRSF21	OCIAD2	CEBPB	HSPA5	SNHG5	GJB3
RPL28	LPCAT4	TXN	PER1	KRT18	DUOXA2
FXYD3	BST2	NR4A1	ADAP1	CA12	HLA-DRB5
SERPING1	EPHA2	AP1M2	TMC4	MYEOV	ARRDC2
HMGA1	MUC5B	SYNPO	HSD17B2	P4HB	OAS1
S100A14	KCNN4	ANO1	PTP4A1	VILL	ADRA2A
H1F0	ITGA3	SIK1	PLAUR	HKDC1	NQO1
KRT17	TSTA3	EFNB2	ANXA10	STEAP3	LRG1
CSTB	IFI27	CDKN1A	PYGB	PLEK2	SERPINH1
ASS1	MMP1	S100A10	CYP2S1	CD68	COL18A1
EFNB1	SLPI	PKP3	ALDOA	MAFF	CERCAM
MDK	CLTB	MCM7	CDCP1	CD9	CA9
LY6E	SPINK1	RARRES3	IFITM3	PLLP	CDHR2
CAMK2N1	JUP	CEACAM1	BAIAP2L1	EPHB4	ESRP1
MGLL	PPP1R15A	ALDH2	MGST1	KRT16	CRYZ
PDZK1IP1	TRIM29	CES2	TFF1	ABLIM3	TAGLN2
BCL2L1	PPP1R1B	SLC6A8	MUC4	PTPRU	HNF1B
C19orf33	ARPC1A	PPL	MMP28	MUC6	BMP4
SFN	GPRC5A	NDRG1	CLDN1	FDFT1	MT1E
ITGB4	RAB25	AZGP1	TM4SF1	GAPDH	ANGPTL4
TSPAN1	ZFP36L2	S100A11	TFF3	EPS8L3	CFB
SLC16A3	CLDN7	SEMA3B	UNC13D	CLDN10	GDF15
C4A	DDIT4	TNFRSF12A	SERINC2	ITPKC	PADI1
KLF5	FKBP4	SOX9	SPNS2	LOXL2	TMSB10
IER2	LAPTM4B	PMEPA1	AQP5	AMIGO2	B2M
ZFP36L1	MUC13	CYR61	IGFBP4	DUOX2	CAV2
MAL2	PSCA	EPCAM	SLC7A5	CLDN4	CTSH
SCNN1A	SDCBP2	BHLHE40	TESC	CAPG	PABPC1
MALL	PROM2	CX3CL1	IFITM1	CEACAM5	KRT8
BGN	LRRC8A	SMAD3	KCNK5	VWA1	TAP1